

Causal Claims with Game-Theoretic Models*

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Abstract

Game-theoretic models support causal arguments, but what follows from a model need not be identified by observed play. I use Dawid's decision-theoretic framework to characterize what an augmented representation of a finite game preserves and develop a protocol for evaluating causal claims. The analysis specifies interventions and separates regime-specific predictions from assumptions connecting the same unit across regimes, and both from what the evidence identifies. Under stated conditions, successful deterrence produces identical recorded play across accounts whose attribution answers span the full probability range. A candidate-entry application shows how accounts fitting the entire record imply sharply different attribution probabilities, and how additional evidence or assumptions can narrow the answer. These limits do not disqualify mechanism-based explanations. The protocol asks researchers to distinguish what the model implies, what supports its mechanism, and what the evidence establishes, while stating what would narrow any remaining uncertainty.

*Version 2; previous versions entitled *The Causal Content of Game-Theoretic Models*.

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1 Introduction

When political scientists explain why conflicts do not occur, incumbents go unchallenged, or coercive demands meet no resistance, they often rely on game-theoretic logic. An aggressor refrains because the defender would respond; a quality challenger stays out because the incumbent would campaign aggressively; a target complies because the sender would follow through (Schelling 1960; Fearon 1995; Powell 1999; Jacobson and Kernell 1983). The response that does the explanatory work is never observed because anticipating it prevents the action that would have called it forth. What must a researcher establish before such a game-theoretic explanation supports a causal claim?

This paper develops and formally grounds a reporting protocol organized around three questions: What is the model? What is the causal question? What is the evidence? The purpose is constructive. An augmented representation supplies a grammar for causal claims about strategic behavior; the analytical results distinguish what the model implies from what observations establish; and the protocol makes those distinctions usable in research.

Dawid’s decision-theoretic approach begins with possible decisions and the distributions of outcomes under each decision (Dawid 2000, 2021), rather than with unit-indexed potential outcomes (Rubin 1974) or structural equations (Pearl 2009). It extends naturally to a game because a strategy is a rule for acting at every information set a player might face. The representation developed here establishes exactly what it preserves for a stated class of finite, perfect-recall extensive-form games. The extensive form supplies moves, information, Nature and terminal consequences; a behavioral profile supplies the players’ rules; together they generate a distribution of play. A solution concept restricts profiles, and selection chooses among those it permits. Prior work connects graphical models and games (Koller and Milch 2003; Hammond et al. 2023; Vonk, Gonzalez Soto, and Kononova 2025); here the primitive object is the extensive form, and the representation separates its contribution from behavior, solution, further assumptions and evidence. Representing the game does not establish its credibility, and a player’s information set does not say what the researcher observes.

A causal question is incomplete until the proposed change is specified. “What if the Challenger entered?” might replace a strategy, force an action, introduce an execution error, reveal a different type, or change the game and solve it again. The same initial move can then produce differ-

ent subsequent behavior. The commensurability problem studied by Bueno de Mesquita and Tyson (2020) explains why an empirical manipulation answers a question about strategic behavior only when it preserves the action’s strategic meaning and consequences for later play. The analysis here also distinguishes asking how outcomes change under a different rule from asking whether a factor caused an observed outcome. These directions are separate from whether the target concerns a population or a particular case, and whether the change concerns external conditions or actors’ choices. Some answers use only the outcome distributions generated separately under each regime; others also require a connection between the same units’ outcomes across regimes. Following Dawid, the paper calls the former sheep and the latter goats. A solved game generally supplies the separate distributions, leaving that connection open.

Useful game-theoretic explanation need not establish what caused one case. A game can isolate a strategic mechanism, and credible comparisons or case evidence can support its relevance without settling particular-case attribution. Section 4.1 connects this distinction to work on models as maps, analytic narratives, and the integration of theory and evidence. The protocol asks that the strength of the evidence match the claim.

Observed play presents a further problem even for causal questions that require no same-case pairing. Accounts of a game that imply the same population distribution of recorded play can give different causal answers. The answers left possible may form a single value, informative bounds, or the full range allowed by the maintained assumptions. Successful deterrence gives the sharp strategic case. In a commitment game, let q be the proportion of Challengers who would enter without commitment. Under the maintained payoff orderings and a rule selecting commitment at indifference, a uniquely selected, sequentially rational profile produces commitment followed by nonentry for every $q \in [0, 1]$. The probability that commitment caused the observed nonentry nevertheless equals q . Exact knowledge of that population record therefore leaves the full range open. Successful deterrence suppresses the behavior needed to learn its effect: strategic behavior helps determine which evidence exists.

The candidate-entry application shows the protocol at work. Full deployment of an incumbent’s campaign resources is uniquely optimal under the maintained assumptions. A record of deployment, every entry decision, and entrant quality still permits two accounts attributing 1/12 and 11/12 of nonentry to the additional resources. A stable challenger type across regimes

supplies the declared same-case connection; the unresolved issue is the distribution of nonentrant types. Additional measurements or a valid resource comparison can narrow attribution. The application shows precisely what independently assigned resources identify and what the further monotonicity assumption contributes.

The contribution lies in connecting three tasks: representing the strategic model with an exact preservation result, specifying the causal target and intervention, and determining which competing answers the evidence distinguishes. The reporting protocol asks researchers to expose the assumptions connecting those tasks and name the evidence or restriction that could resolve remaining disagreement. An exact answer, informative bounds, or the full range may each be defensible; readers need to see what supports the reported conclusion.

Sections 2–4 develop the model, causal-question and evidence arguments in that order. Section 5 applies them to candidate entry, including a better empirical comparison and a completed protocol. Section 6 draws the implications for causal claims made with games.

2 What Is the Model?

A game specifies the structure of possible play; behavioral rules say how players act within it. Dawid’s decision-theoretic framework makes this distinction explicit by representing outcome distributions under specified decision rules (Dawid 2000). The task here is to represent an extensive-form game, identify what that representation preserves, and then show what strategies and a solution add.¹

2.1 From One Decision to Interdependent Decisions

Dawid’s decision-theoretic approach starts with choosing an action by comparing its possible consequences and the losses attached to them. Consider a person with a headache deciding whether to take aspirin. Let Y be headache

1. Influence diagrams represent decisions under uncertainty (Howard and Matheson 2005; Dawid 2002); multi-agent influence diagrams and subsequent graphical game models extend that approach to strategic settings (Koller and Milch 2003; Hammond et al. 2023; Vonk, Gonzalez Soto, and Kononova 2025). Here the primitive object is the extensive-form game.

duration and $L(Y)$ the loss the person attaches to that duration. Suppose an experiment supplies predictive distributions P_t under aspirin and P_c under control for people like the decision maker. The person can choose the action with the lower expected loss by comparing $\mathbb{E}_{P_t}[L(Y)]$ and $\mathbb{E}_{P_c}[L(Y)]$. This comparison uses the two distributions separately; it requires no joint distribution of the same person's duration under both choices.

A *regime* g is a rule assigning an action, possibly using information available at the decision. Write $\mathcal{G}$ for the regimes considered and $P_g(Y)$ for the outcome distribution, or law, under g . Their collection $\{P_g(Y) : g \in \mathcal{G}\}$ is Dawid's *physical model*. An *effects-of-causes* question compares features of these laws, such as

$$\mathbb{E}_{g_1}[Y] - \mathbb{E}_{g_0}[Y]. \quad (1)$$

The expectation's subscript names the regime or profile supplying its distribution. In a game, several actors' rules jointly determine these distributions.

2.2 From a Decision to a Game

In Figure 1, a Challenger, C , decides whether to enter an electoral contest. Nonentry ends the game; entry gives the Incumbent, I , a choice between fighting and accommodating. Nonentry pays $(0, m)$, fought entry $(-c, -c_F)$, and accommodated entry (π_C, π_I) . Assume $\pi_C > 0 > -c$ and $-c_F < \pi_I < m$. The Challenger prefers accommodation to nonentry to fighting; the Incumbent prefers nonentry to accommodation to fighting.

The decision variables record the moves: D_C takes *Enter* or *Stay Out*; D_I takes *Fight*, *Accommodate*, or $\emptyset$ when the Incumbent's decision is not reached. The terminal variable Z identifies the completed path, from which payoffs and other outcomes follow. The arrow $D_C \rightarrow D_I$ records ancestry and possible activation of the Incumbent's decision. That the Incumbent observes entry comes from its information set, not from the arrow alone: an earlier move can precede another player's choice without being observed. Arrows into Z record the dependence of completed play on the moves that occur.

Neither panel supplies a behavioral rule for the Incumbent's response. Unlike the aspirin choice followed by chance, entry leads to another actor's decision. Only after rules are assigned to both players does the model generate a distribution of play and hence expected payoffs. Maximizing those payoffs is the counterpart of minimizing losses, with $u = -L$. Observed

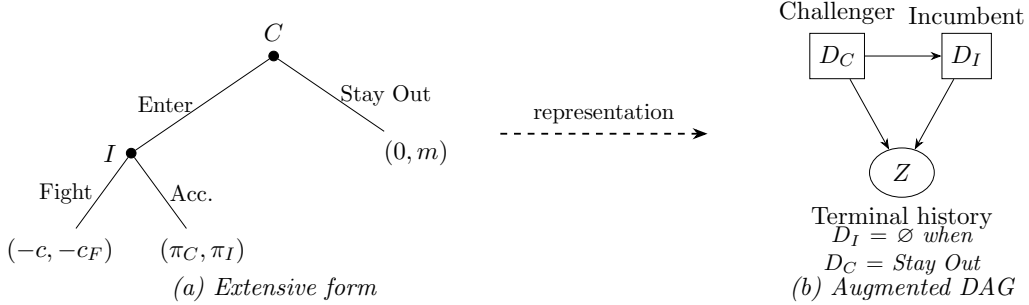


Figure 1: The entry-deterrence game as an extensive form and an augmented DAG. Squares are decision variables, while the ellipse contains Z , the terminal-history variable. The symbol $\emptyset$ records that the Incumbent’s decision was not reached. The players’ payoffs are functions of the completed path and are not drawn as separate nodes.

nonentry does not itself reveal the unreached response.

2.3 What the Extensive Form Contributes

In a finite extensive-form game Γ , a rooted tree specifies possible histories, assigns each move to a player or Nature, and attaches payoffs to terminal histories. An information set groups decision nodes that its owner cannot distinguish and gives them the same available actions. Let N denote the players.

The representation uses one decision variable D_h per information set h and one chance variable W_t per Nature node t . Their pointed domains consist of the available actions plus a distinguished nonreachability value $\emptyset$, which is not an action. Each complete play yields a *structural assignment*: the action at every reached move, $\emptyset$ elsewhere, and the terminal identity Z . A player’s payoff $u_i(Z)$ and the substantive outcome $Y = \psi(Z)$ are functions of that identity. Perfect recall—remembering one’s own earlier actions and information—ensures that no play visits an information set twice, so the assignment is well defined.

This construction carries the following features of the extensive form into the variable representation: the typed decision, chance and terminal variables; decision owners and domains with $\emptyset$ distinguished for move variables; the complete set of structural assignments; Nature’s nodewise action prob-

abilities; terminal identities and payoffs. Write $\text{Inv}(\Gamma)$ for this collection. An isomorphism preserves each component, including player identities and $\emptyset$. The augmented representation $\text{Aug}(\Gamma)$ adds literal ancestry arrows: one move points to another if some node represented by the first strictly precedes some node represented by the second, and every move points to Z . Definition 2 in Appendix A.1 gives the formal construction.

Call a game *layered* when one linear order of its information sets and Nature nodes agrees with their order on every play. Its literal move graph is acyclic exactly when it is layered, as the entry game is. Perfect recall alone does not ensure this: pooled information sets can place one decision before another on one path and after it on another.

Some printed order is merely descriptive. Suppose one player chooses and, after every action, a different player chooses at one common information set without observing the first choice. Reversing this adjacent two-move block while preserving every continuation is an *elementary descriptive interchange*. A *labeled isomorphism* renames nodes, actions, information sets, Nature nodes and terminal identities while preserving owners, relations, probabilities and payoffs. Write $\Gamma \sim \Gamma'$ for equivalence generated by these isomorphisms and finitely many descriptive interchanges.

Proposition 1 (Structural representation). *Let Γ and Γ' be finite perfect-recall extensive-form games in which every information set and Nature node has at least two actions.*

- (a) (Information.) *The typed variables and structural assignments in $\text{Inv}(\Gamma)$ carry the game's information sets, their owners and actions, and the paths on which they are reached. A rule at D_h is therefore one rule for the entire information set h , not a rule reconstructed from the arrows.*
- (b) (Strategies.) *A behavioral strategy for player i —one probability distribution over available actions at each of the player's information sets—is in one-to-one correspondence with a family of rules for that player's decision variables, with $\emptyset$ recorded whenever a variable is not reached.*
- (c) (Representation up to descriptive interchange.)

$$\Gamma \sim \Gamma' \iff \text{Inv}(\Gamma) \cong \text{Inv}(\Gamma').$$

The isomorphism preserves the typed variables, their pointed domains, structural assignments, nodewise Nature probabilities, terminal identities, and payoffs. Chance variables are never crossed by an interchange;

their placement is fixed by $\text{Inv}(\Gamma)$, up to the displayed isomorphism. A descriptive interchange can alter only the two possible literal ancestry arrows between the interchanged decision variables, so the raw edge set is not itself invariant. For layered representatives, the literal presentation is a DAG.

Parts (a) and (b) record the construction’s correspondences; part (c) establishes completeness. Within its stated class, the retained information forgets the descriptive order of adjacent unobserved choices and nothing else. The representation is invertible on its image once those encodings are treated as equivalent. The two-action requirement excludes points without branching; it does not require positive probability for every action. The proof and boundary counterexamples are in Appendix [A.1](#).

For this theorem’s chance-free subclass, Bonanno ([1992](#)) establishes a related characterization using interchanges of contiguous simultaneous moves. The present representation additionally retains typed chance variables, their nodewise probabilities and placement, and terminal payoffs. This matters for information-dependent regimes: Nature may generate a signal before a decision, and the information sets specify whether the player’s rule can use it. The appendix distinguishes this equivalence from broader transformations preserving the reduced normal form.

2.4 What Strategies and Solutions Add

A *behavioral strategy* σ_i assigns an action distribution at each of player i ’s information sets (Kuhn [1953](#)), including those the selected play never reaches (Rubinstein [1991](#)). A pure strategy chooses an action at each information set; a mixed strategy randomizes over complete plans. Behavioral strategies instead attach distributions directly to the local decision variables. One for each player gives the profile $\sigma = (\sigma_i)_{i \in N}$.

In the entry game, the profile consists of $\sigma_C(\textit{Enter})$ and $\sigma_I(\textit{Fight} \mid \textit{Enter})$. The latter remains specified even if the Challenger never enters. Its vertical bar labels the decision where the rule applies, not a conditional frequency inferred from observed play (Signorino [2002](#)). The profile produces the path

probabilities

$$\begin{aligned}
P_\sigma(Z = \textit{Stay Out}) &= 1 - \sigma_C(\textit{Enter}), \\
P_\sigma(Z = (\textit{Enter}, \textit{Fight})) &= \sigma_C(\textit{Enter})\sigma_I(\textit{Fight} \mid \textit{Enter}), \\
P_\sigma(Z = (\textit{Enter}, \textit{Accommodate})) &= \sigma_C(\textit{Enter})[1 - \sigma_I(\textit{Fight} \mid \textit{Enter})]. \quad (2)
\end{aligned}$$

These also determine expected payoffs because payoffs are functions of the completed path.

A *solution concept* restricts admissible profiles to a set $\mathcal{S}(\Gamma)$. Unless a selection rule chooses one, the model supplies the family $\{P_\sigma : \sigma \in \mathcal{S}(\Gamma)\}$. In the entry game, subgame-perfect equilibrium uses the Incumbent's strict preference for accommodation and selects $(\sigma_C(\textit{Enter}), \sigma_I(\textit{Fight} \mid \textit{Enter})) = (1, 0)$: entry followed by accommodation. Nash equilibrium also admits $(0, 1)$: nonentry with an unreached prescription to fight. These produce different distributions of play, so a claim about "the Nash equilibrium" must state a selection or carry the alternatives forward.

An *adapted regime* g_i supplies one action distribution at each of player i 's information sets when reached, recording $\emptyset$ otherwise. Replacing σ_i with g_i while holding the other players' rules σ_{-i} fixed therefore defines another behavioral profile. As g_i varies, the induced outcome laws form the game's counterpart of the physical model. This replaces a player's rule; it does not force one realized action or change the game and solve it again.

Proposition 2 (Profile-induced regime laws). *Let Γ be a finite extensive-form game with perfect recall.*

- (a) (From a profile to a distribution.) *Every behavioral profile σ , together with Nature's nodewise probability rules, determines a unique distribution over terminal histories and hence a unique law P_σ over structural assignments, each of which includes its terminal identity. For any outcome $Y = \psi(Z)$, it therefore determines the outcome distribution $P_\sigma(Y)$.*
- (b) (Changing one player's regime.) *Replacing σ_i by an adapted regime g_i gives another behavioral profile (g_i, σ_{-i}) and hence another well-defined distribution $P_{(g_i, \sigma_{-i})}$.*
- (c) (What the replacement holds fixed.) *Under a unilateral change from g_i to g'_i , Nature's nodewise probability rules, every other player's behavioral rules, and the terminal and outcome maps remain unchanged. The*

change can alter which information sets and Nature nodes are reached, and hence the distribution of play. The probabilities assigned at each unchanged point stay fixed; how realized draws are paired across regimes remains unspecified.

Part (c) specifies the comparison: changing the Challenger’s entry rule can bring the Incumbent’s existing response into play, but does not replace that response rule with a newly selected one.

The proof in Appendix A.2 recursively multiplies action probabilities along the tree. It requires no DAG; only for layered games does the resulting law also factorize over the literal augmented DAG. Thus probabilities over play belong to (Γ, σ) , not to the extensive form alone.

2.5 What the Representation Does Not Establish

The representation preserves the proposed actors, options, information and incentives; it does not test them. An ancestry arrow does not establish statistical dependence, and an information set describes a player’s knowledge rather than the researcher’s record. Likewise, a solution does not establish that players follow its rules or hold its beliefs. Even a unique solution is a conditional prediction.

Nonentry might agree with a deterrent equilibrium without showing that the Incumbent would fight, that the Challenger stayed out for that reason, or that the game includes the relevant considerations. The next section specifies the causal claims those rules could support. Section 4.1 then distinguishes evidence supporting the mechanism from evidence establishing attribution.

3 What Is the Causal Question?

The game, profile, and Nature’s probabilities generate an outcome distribution under each specified regime. Those distributions answer questions about the consequences of a change, conditional on the model. Attributing an observed outcome to that change may also require knowing how the same case would fare under both regimes. Before asking whether the model supplies that connection, the researcher must state the target and the change: a new strategy, a forced action, and a new game can imply different causal answers.

3.1 What Kind of Causal Question Is Being Asked?

Section 2.1 introduced an effects-of-causes question: how would the distribution of an outcome change if the decision rule changed? In the entry game, for example, one can compare conflict under a rule that makes the Challenger enter with conflict under a rule that makes the Challenger stay out, while holding the Incumbent’s response rule fixed. One can instead begin with a contest in which the Challenger stayed out and ask whether the threat of a fight caused that nonentry. The first question runs from a specified change to its possible effects; the second begins with an observed outcome and asks whether it should be attributed to a cause.

Questions of the second kind are called *causes-of-effects* questions. To state one, let Y^g denote the outcome that a case would have under regime g . This notation does not by itself supply a joint distribution for outcomes under two mutually exclusive regimes. For a binary outcome observed as $Y = 1$ under g , the *probability of causation* asks for

$$P\left(Y^{g'} = 0 \mid Y^g = 1\right),$$

the probability that the outcome would have been absent under the alternative regime g' . Unlike a comparison of $P_g(Y)$ and $P_{g'}(Y)$, this question refers to outcomes for the same cases under both regimes.

The direction of the question is distinct from whether the answer concerns one case or a population. An effects-of-causes question can be tailored to an actor with particular characteristics, while a causes-of-effects probability can be reported for a class of cases sharing the same evidence. Nor does the distinction say what has been changed: an external feature of the situation, a player’s decision rule, and one externally forced action pose different questions even when they produce the same immediate move.

The questions also differ in the ingredients their answers require. Following Dawid (2000), call a question a *sheep* when its answer depends only on the regime-specific outcome distributions that Section 2.1 called the physical model, and a *goat* when it also depends on how outcomes under mutually exclusive regimes are related. The average treatment effect represented by Equation 1 is a sheep because it uses one expectation from each of two regime-specific distributions. A causes-of-effects attribution is generally a goat, although additional structure may make a particular attribution depend only on a regime-specific distribution. Sheep and goat are therefore

not alternative names for effects of causes and causes of effects, for population and particular-case questions, or for interventions on external conditions and players' choices. They also do not say what observed evidence reveals: observed play may leave a sheep unanswered, while additional evidence or assumptions may settle a goat.

3.2 What Exactly Is Being Changed?

The notation Y^g is useful only after the alternative regime g has been specified. In a game, a counterfactual phrased only as an action does not yet do that. If a selected profile assigns entry probability zero, “what if the Challenger had entered?” says which move differs but not why it differs or what else remains fixed. Two choices must be separated: what is changed, and whether the remaining behavioral rules are held fixed or selected anew.

One possibility is the strategy replacement defined in Section 2.4. Starting from the deterrent Nash profile, replace the Challenger's rule with a rule that always enters while holding the Incumbent's rule fixed. The Incumbent's unreached prescription remains *Fight*, so the replacement produces fought entry under Proposition 2; the resulting profile is not claimed to be an equilibrium. This is a strategy intervention because it replaces the Challenger's complete rule, including prescriptions at every information set the Challenger might face. The Challenger has only one information set in this game, but that simplification disappears in larger games.

An external device might instead force *Enter* at the Challenger's decision while leaving the Challenger's prescribed strategy in place. In this simple game, forcing entry whenever the initial decision is reached can produce the same distribution of play as the always-enter strategy replacement when every other rule is held fixed. The causal interpretations nevertheless differ: one changes the player's rule, while the other overrides the action that the rule would have produced. In a larger game, moreover, replacing a complete strategy governs several possible decisions, while a forced-action intervention may alter only one. For the same reason, an empirical estimate obtained by forcing an action answers a question about replacing a player's strategy only when the manipulation preserves the action's strategic meaning and its consequences for subsequent behavior (Bueno de Mesquita and Tyson 2020). Conditioning on cases in which entry happened—possible only under rules that give entry positive probability—is different from either intervention because it changes no rule and forces no action.

The alternative may instead change the strategic situation itself. A different payoff, action set, move order, information structure, or commitment technology changes some part of the game; a different player type may be a realization already represented by Nature or may require adding something the game omitted. Such a change does not by itself specify a new strategy profile. Holding the old behavioral rules fixed, where they remain well defined in the changed game, answers one question, while selecting a solution of the changed game answers another and may alter several players' behavior.

A mistake or tremble also needs interpretation. A one-time execution error overrides an intended action, whereas a behavioral rule with full support gives every available action positive probability whenever its information set is reached. Quantal-response equilibrium is one familiar model of such stochastic choice, with choice probabilities responsive to expected payoffs (McKelvey and Palfrey 1995). A refinement adds requirements used to distinguish among solutions: trembling-hand perfection asks which equilibria remain robust when unintended actions have small positive probability, while sequential equilibrium makes explicit the beliefs players hold at information sets, including unreached ones, and disciplines them (Selten 1975; Kreps and Wilson 1982). A payoff or type perturbation instead changes the players' incentives or information (Harsanyi 1973). These devices can make an off-path question precise, but they need not define the same alternative or imply the same continuation. In the entry game, the readings diverge downstream: with the deterrent rules held fixed, a forced or mistaken entry is met with *Fight*, while entry understood as evidence of a different game or Challenger type, followed by a new solution, can end in *Accommodate*.

Stating a counterfactual this fully is the part of the closest-world problem that matters here (Lewis 1973; Fearon 1991). No general similarity metric is required, but the causal question must state what changes, what remains fixed, and whether the game is solved again. Choosing among those specifications is a substantive part of the causal claim and must be defended for the application rather than treated as a technical convenience. Only then can the analysis ask what the resulting outcome distributions do and do not determine.

3.3 What Does a Solved Game Determine?

Once a profile has been selected and an alternative has been specified, Proposition 2 supplies a separate outcome distribution under each regime. Write

σ_{-i}^* for the maintained rules of the players other than the focal player i , whether selected by a solution concept or simply held fixed. For brevity, write μ_r for the distribution of Y under regime r , those maintained rules, and Nature’s probability rules. An effects-of-causes question can compare μ_g and $\mu_{g'}$ directly. The probability of causation in Section 3.1 asks for something else: how the outcomes under the two regimes are paired for the same cases.

A *coupling* of μ_g and $\mu_{g'}$ is a joint distribution for $(Y^g, Y^{g'})$ with those two margins. It says which outcome under g is paired with which outcome under g' for the same underlying case.

A one-decision binary game makes the omission visible. Let $A \in \{0, 1\}$ record the action taken at the single decision, call $A = 1$ a success, and let the outcome be $Y = A$. Suppose $P_g(A = 1) = 1/4$ while $P_{g'}(A = 1) = 3/4$. Scale those probabilities to a population of 100 cases, and suppose g' is the regime actually observed; the probability of causation then asks for $P(Y^g = 0 \mid Y^{g'} = 1)$. One possible pairing makes the 25 successes under g a subset of the 75 successes under g' . On that pairing, 50 cases change their outcomes, and 50 of the 75 observed successes would have failed under g . The probability of causation is therefore $2/3$. Another pairing matches the 25 successes under g with the 25 failures under g' and the 75 failures under g with the 75 successes under g' . Every case then changes its outcome, and every observed success would have failed under g , so the probability of causation is one. These are the lowest and highest values permitted by the two regime distributions—the extremes attainable under any joint distribution with those two margins. The regime distributions can therefore be known exactly while leaving a consequential attribution unsettled.

Further assumptions can select or restrict the missing pairing, but those assumptions do not come from the two regime distributions alone. If no case can change from success under g to failure under g' , so that $A^{g'} \geq A^g$ for every case, *unit-level monotonicity* selects the first pairing and its probability of causation of $2/3$. The ordering $3/4 > 1/4$ is necessary for that claim but does not establish it (Tian and Pearl 2000; Dawid and Musio 2022).

The example pairs outcomes directly; a game requires the pairing to respect movement through its tree. A *compatible structural implementation* $\mathcal{I}$ collects the random inputs used to generate Nature’s moves and players’ randomized actions in a random element ξ . Its *law* P_Ξ is the probability distribution of those inputs. Both regime calculations use the same realized inputs; the implementation specifies how they translate into actions. For each

regime it supplies a deterministic measurable map from ξ to an action at every information set and Nature node, reached or not; pure prescriptions use constant maps. Following these maps through the tree must reproduce each regime's full profile-induced distribution of play. At every unchanged same locus, including unchanged prescriptions of the focal player, the maps agree P_Ξ -almost surely. Only that player's loci whose prescriptions change may use different maps across regimes. Maps at distinct loci on regime-separated branches need not agree. Appendix A.3 gives the formal definition.

In the binary example, the two pairings arise from implementations that differ only in the alternative-regime map at the focal decision whose prescription changes, so both satisfy the unchanged-rule requirement. A common source of randomness is not by itself enough to select between them; the maps from that source to actions do the selecting.

Once an implementation has been specified, its common random element and the regime together determine the completed path and its outcome. Dawid calls a random element that performs this role a *determining concomitant* (Dawid 2000, 2021).

Definition 1 (Determining concomitant). *Let $\mathcal{R}$ be a family of regimes. A regime-invariant random element η is a determining concomitant for an outcome Y over $\mathcal{R}$ if there are measurable maps f_r such that*

$$Y^r = f_r(\eta) \quad \text{almost surely for every } r \in \mathcal{R}.$$

Regime invariance means that the same realized value of η is used under every regime, not merely that η has the same distribution under each one. It does not require η to occur earlier in the game's clock time.

Lemma 1 (What regime-specific laws determine). *Let Γ be finite with perfect recall, fix player i and σ_{-i}^* , and let $Y = \psi(Z)$ be a specified terminal-history outcome. Consider two regimes g_i, g'_i for player i .*

- (a) *For each $r \in \{g_i, g'_i\}$, the profile (r, σ_{-i}^*) and Nature's nodewise probability rules determine a unique outcome distribution $\mu_r := P_{(r, \sigma_{-i}^*)}(Y)$. Every target defined solely as a function of $(\mu_{g_i}, \mu_{g'_i})$ is therefore determined without a cross-regime joint distribution.*
- (b) *Every compatible structural implementation $\mathcal{I}$ determines, for each $r \in \{g_i, g'_i\}$, a measurable map*

$$F_r^{\mathcal{I}}(\xi) := \psi(z^{\mathcal{I}}(r; \sigma_{-i}^*, \xi))$$

such that $Y^r = F_r^{\mathcal{I}}(\xi)$ and

$$\mathcal{L}_{\mathcal{I}}(Y^{g_i}, Y^{g'_i}) = (F_{g_i}^{\mathcal{I}}, F_{g'_i}^{\mathcal{I}})_{\#} P_{\Xi}.$$

Here $z^{\mathcal{I}}(r; \sigma_{-i}^*, \xi)$ is the terminal history reached by following the implementation's maps under regime r at the realization ξ . The joint distribution is obtained by applying the two displayed maps to the same ξ distributed according to P_{Ξ} . The implementation thus supplies a coupling of μ_{g_i} and $\mu_{g'_i}$, and ξ is a determining concomitant. Once $\mathcal{I}$ is fixed, no additional coupling choice is required.

- (c) *The behavioral rules and Nature's nodewise probability rules alone do not generally determine $\mathcal{L}(Y^{g_i}, Y^{g'_i})$. Distinct compatible implementations can reproduce the same full distribution of play within each regime and induce different joint distributions across regimes.*

The lemma applies to outcomes represented as functions of the game's terminal history. A causal question about a feature absent from that representation requires the model to be augmented before the result applies.

Fixing a stochastic behavioral rule fixes its action probabilities whenever the relevant information set is reached. It does not say how the realized random action in one regime is paired with the realized action in another. A pure strategy removes that uncertainty only at the decisions governed by that strategy; Nature's moves and other players' randomization may still leave the cross-regime pairing open. If the outcomes under both regimes are completely deterministic, by contrast, there is only one possible pairing.

For a target that depends on the pairing, the analysis must therefore state or restrict a compatible implementation, or show that the target has the same value under every compatible implementation. Targets that depend only on the separate regime distributions satisfy the second condition automatically. The target of the Deterrence Corollary in Section 4 has that property for a different reason. When the factual event $Y^g = 1$ has probability one,

$$P(Y^{g'} = 0 \mid Y^g = 1) = P(Y^{g'} = 0).$$

Its probability of causation therefore reduces to the probability, under the alternative regime alone, that the outcome would have been absent, so the pairing is irrelevant to that target even though it may matter for others.

Appendix A.3 proves the lemma and gives the boundary examples, a sufficient construction, and the qualifications for comparisons involving more than two regimes.

4 What Is the Evidence?

A game may yield an exact causal answer without evidence that its assumptions describe the application. Evidence can support the proposed mechanism and can distinguish accounts that give different causal answers. These contributions should be assessed separately.

4.1 What Can the Game Explain?

Given the game, the players' rules, and Nature's probabilities, Proposition 2 determines the distribution of play under each specified regime. Comparing those distributions shows how incentives, information, and expectations produce behavior inside the model. This is a conditional implication: if the stated actors, options, information, incentives, and rules are right, the consequence follows.

Evidence can support a further claim about the mechanism. Suppose credible empirical comparisons show that behavior changes as the game predicts when the corresponding incentives or information change. The empirical change and outcome must correspond to those in the model, and the features treated as unchanged must be defensible. Agreement then gives reason to believe that the game captures how actors respond, especially when other evidence supports its account of their options, information, and incentives. It does not establish that the game is the only explanation or that every unobserved part is correct. Work on empirical implications of theoretical models and the integration of theory and evidence emphasizes both the value of these tests and the arguments connecting empirical comparisons to theoretical quantities (Granato and Scioli 2004; Aldrich, Alt, and Lupia 2008; Ashworth, Berry, and Bueno de Mesquita 2021).

That middle claim is a legitimate form of explanation. Models can function as maps or fables that isolate a mechanism without reproducing every feature of the world (Clarke and Primo 2012; Rubinstein 2012; Johnson 2021). Analytic narratives combine strategic models with detailed case evidence to argue that an outcome arose for reasons represented in the model; debate over that practice shows why the supporting evidence must be explicit (Bates et al. 1998, 2000; Elster 2000). The explanation draws its force from the argued connection between model and application, not from formal structure alone.

The claim that a threat caused a particular nonentry goes farther: it says

how the same contest would have ended without the threat. Agreement with comparative-static predictions does not by itself supply that connection. The stronger claim requires the structure described in Section 3.3, or a demonstration that the target does not depend on the structure left open. A game can therefore contribute to causal explanation without settling particular-case attribution. The next question is which causal answers observed play can distinguish.

4.2 When the Same Observations Support Different Answers

Return to the entry game from Section 2.2. Suppose the researcher records only whether the Incumbent fights, grouping accommodated entry and nonentry together. The accommodating Nash equilibrium produces entry followed by accommodation; the deterrent equilibrium produces nonentry with an unreachd prescription to fight. Both generate the same population record: no fighting with probability one.

Replace the Challenger's strategy with always entering, holding the Incumbent's strategy fixed. The accommodating account then predicts no fighting; the deterrent account predicts fighting with probability one. The game, Nash solution concept, replacement and recorded population distribution are identical; only equilibrium selection differs. This effects-of-causes question requires no cross-regime pairing, yet the observed record leaves its answer unsettled. Recording entry as well would distinguish the accounts. More observations of fighting alone, drawn from the original population, would not.

To separate this problem from sampling uncertainty, suppose the population distribution of the recorded variables is known exactly. Accounts can fit that distribution equally well while giving different causal answers. The following construction collects the answers that survive.

Fix the game form of a finite, perfect-recall game: its tree of possible moves, the player or Nature assigned to each move, its available actions and information sets, and its terminal histories, but not yet its payoff values or Nature's probabilities. Also fix a solution concept, a causal question, and a rule specifying what the researcher records. A *complete account*, denoted by M , supplies the payoffs, Nature's probabilities, and one behavioral strategy profile permitted by the solution concept. It also supplies a compatible

structural implementation when the causal question requires the same-case connection discussed in Section 3.3. In that event, the implementation's regime-invariant random element is a determining concomitant in the sense of Definition 1. The maintained assumptions determine which complete accounts are allowed; let $\mathcal{M}$ denote the set of allowed complete accounts.

Let O denote the rule that turns completed play into the variables the researcher records. In the entry example, O reports whether fighting occurred but not whether the Challenger entered. For each account $M \in \mathcal{M}$, let P_O^M be the population distribution of those recorded variables generated by its selected strategy profile. Let P_O° be the population distribution actually observed. The accounts that reproduce the observations are

$$\mathcal{M}^\circ = \{M \in \mathcal{M} : P_O^M = P_O^\circ\}.$$

Thus $\mathcal{M}^\circ$ is a set: it contains every allowed complete account that produces the same population distribution of recorded play.

The causal question must also be held constant across those accounts. Let $\tau(M)$ denote its answer under account M . The answers left possible by the maintained assumptions and the observed play form the set

$$\mathbb{I}_\tau = \{\tau(M) : M \in \mathcal{M}^\circ\}.$$

This set may contain one answer, restrict an ordered answer to informative bounds, or contain every value allowed by the maintained class. Structural parameters can be identified despite equilibrium multiplicity (Tamer 2003); the object here is the set of causal answers compatible with the maintained assumptions and recorded play.

Proposition 3 (What observed play can distinguish). *Assume $\mathcal{M}^\circ \neq \emptyset$.*

- (i) *Observed play singles out one answer within the maintained class if and only if*

$$\tau(M) = \tau(M') \quad \text{for every } M, M' \in \mathcal{M}^\circ.$$

- (ii-a) *Once $(\mathcal{M}, O, P_O^\circ, \tau)$ is fixed, so is $\mathbb{I}_\tau$. Additional observations of the variables recorded by O , drawn from the same population distribution, can improve finite-sample precision but cannot change this population set.*

(ii-b) If $M, M' \in \mathcal{M}^\circ$ and $\tau(M) \neq \tau(M')$, both answers belong to $\mathbb{I}_\tau$. Every sample has the same distribution under the two accounts, so observations drawn from P_O° cannot distinguish them. Let $P_{d,O}^M$ denote the population distribution of the variables recorded by O under an additional specified regime d and account M . Population observations under d distinguish the pair exactly when

$$P_{d,O}^M \neq P_{d,O}^{M'}.$$

Narrowing $\mathbb{I}_\tau$ requires excluding at least one account that currently survives. A stronger maintained assumption can do so, as can evidence whose distribution differs across those accounts. The two sources should not be confused: an assumption may settle the answer without having been learned from observed play. Conversely, allowing payoffs, Nature's probabilities, the selected profile, or the implementation to vary can widen $\mathbb{I}_\tau$ only if the permitted variation changes τ . Appendix A.5 states the qualifications needed when several ingredients cannot vary independently and gives a template for demonstrating which difference between two accounts is responsible for their different answers.

The set $\mathbb{I}_\tau$ records disagreement among accounts fitting the population evidence, whereas a confidence interval represents finite-sample uncertainty. A singleton does not establish that the maintained game and behavioral assumptions describe the world correctly. It says that every permitted account consistent with the record gives the same answer. The next example shows that disagreement can survive richer recording and unique, sequentially rational selection.

4.3 How Deterrence Can Hide Its Own Evidence

How can observed play leave the full range open even when the researcher records both commitment and nonentry and the game has a uniquely selected, sequentially rational profile? Strategic behavior can suppress the evidence needed to answer the causal question. A commitment that prevents every challenge also prevents the researcher from observing how challengers would have behaved without it.

Consider a population of contests described by a richer entry game. The Incumbent first chooses whether to make a hands-tying commitment. Let $X = 1$ denote commitment and $X = 0$ its absence. Nature then draws the

Challenger's type $V \in \{L, H\}$, with

$$q = P(V = H).$$

The players know q and observe both the commitment choice and the realized type, but the researcher does not know q and does not observe V . The Challenger then chooses whether to enter. Following entry, the Incumbent chooses whether to fight or accommodate. Let $Y = 1$ denote nonentry and $Y = 0$ entry. The researcher records $O = (X, Y)$: whether the Incumbent committed and whether the Challenger entered. The conclusion below depends on that record; adding V to the population record would reveal q .

Assume that the payoffs make an uncommitted Incumbent accommodate entry and a committed Incumbent fight. Commitment has no direct payoff effect when entry does not occur, and the Incumbent prefers uncontested office to accommodating entry. Both Challenger types prefer nonentry to entering against a fight. Against accommodation, the high type H enters and the low type L stays out. It follows that H enters only when $X = 0$, whereas L never enters. Commitment is strictly optimal for the Incumbent whenever $q > 0$ and is tied with noncommitment when $q = 0$. The maintained selection rule chooses commitment at that tie. Together with the strict continuation choices, this rule selects a unique sequentially rational profile for every $q \in [0, 1]$. These payoff and selection assumptions are maintained rather than inferred from the observed peace; Appendix A.7 gives the inequalities and derivation.

Write σ^* for the selected profile. Write $P_{\sigma^*, O}$ for the population distribution of the recorded variables under that profile. Under σ^* , the Incumbent commits and both Challenger types stay out. Hence

$$P_{\sigma^*, O}(X = 1, Y = 1) = 1$$

for every value of q . Even exact knowledge of the population distribution of (X, Y) therefore reveals nothing about the proportion of high types. Requiring every action to occur with positive probability changes the behavioral account rather than extracting more information from this population record. A full-support model such as quantal-response equilibrium predicts some entry and therefore a different population distribution of (X, Y) ; trembling-hand perfection instead tests the limiting profile's robustness and need not give entry positive probability in the limit (McKelvey and Palfrey 1995; Selten 1975).²

2. I thank an anonymous referee for prompting this clarification.

The causal question asks how much of the observed nonentry was due to commitment. More precisely, among contests with commitment and nonentry, what proportion would instead have ended in entry if the Incumbent's commitment rule had been replaced by $X = 0$? Let $Y^{X=0}$ denote the outcome under that replacement, which holds every subsequent strategy prescription fixed. In particular, the uncommitted Incumbent still accommodates entry. The replacement defines the causal comparison; it is not a claim that the modified profile is another equilibrium. The probability of causation (Dawid, Musio, and Fienberg 2016) is

$$\text{PC} = P(Y^{X=0} = 0 \mid X = 1, Y = 1).$$

Because $(X, Y) = (1, 1)$ has probability one, the conditioning does no work. Under the replacement, H enters and L stays out, so $\text{PC} = q$. Because $Y = 1$ for every observed contest, this target depends only on the distribution of $Y^{X=0}$, not on a joint distribution connecting the factual and alternative outcomes for the same contests.

Corollary 1 (Deterrence can leave the full range open). *In the deterrence game just described, maintain sequential rationality and the selection rule that chooses commitment when $q = 0$. For every $q \in [0, 1]$,*

$$P_{\sigma^*, O}(X = 1, Y = 1) = 1 \quad \text{and} \quad \text{PC} = q.$$

Because the recorded population distribution is the same for every q ,

$$\mathbb{I}_{\text{PC}} = [0, 1].$$

If the selection rule instead uniquely chooses noncommitment when $q = 0$, the observed commitment excludes that endpoint and $\mathbb{I}_{\text{PC}} = (0, 1]$. If the maintained class admits every equilibrium selection allowed by the solution concept, its committing selection at $q = 0$ restores $\mathbb{I}_{\text{PC}} = [0, 1]$. If independent information restricts $q \in [a, b]$, the corresponding set is $[a, b]$ under a rule that selects commitment at indifference or under all allowed selections. When $a > 0$, every selection treatment gives $[a, b]$. If $a = 0 < b$ and the unique selection rule chooses noncommitment at $q = 0$, the set is $(0, b]$; if $a = b = 0$, that rule is inconsistent with the observed commitment. These conclusions maintain the response and payoff orderings just stated. If those orderings are also allowed to vary, additional response patterns become possible, and the union of the answer sets they allow, with q unrestricted, is still $[0, 1]$.

The corollary gives the strongest failure possible: observed play rules out no answer. This is not uncertainty caused by a small sample. Every additional contest generated by the same selected profile records commitment and nonentry again. Nor is commitment assigned independently of the strategic setting. The Incumbent chooses it because it deters entry, and its success keeps the behavior needed to assess its effect out of view. The strategic behavior under evaluation therefore determines which evidence exists.

Two parts of the game go unobserved, and they matter for different claims. Because $X = 0$ never occurs, the researcher never observes the Challengers' decisions without commitment. That absence conceals q and therefore leaves the probability of causation open. Because entry never follows $X = 1$, the researcher also never observes whether the committed Incumbent would actually fight. That absence bears on the credibility of the threat, but it does not widen these particular bounds: the causal comparison passes through the $X = 0$ branch, where the maintained strategy prescribes accommodation. The observed peace supplies direct evidence for neither response, but the two missing responses answer different questions.

The next subsection asks what additional assumptions or evidence could narrow the range and which part of the argument each addition would support.

4.4 What Additional Evidence Can Do

In the deterrence example, evidence narrows attribution by restricting q . Evidence about payoffs or behavior instead supports or challenges the assumptions making $PC = q$. These are distinct contributions.

Independent information about the Challenger population can bound the answer. If measurements taken before entry, or evidence from comparable contests, restrict the proportion of high types to between one-quarter and three-quarters, Corollary 1 gives

$$\mathbb{I}_{PC} = [0.25, 0.75].$$

Those bounds come from the additional type information. Support for the payoff, behavioral and selection assumptions justifies translating that information into attribution but does not itself reveal q . Recording V for the original contests would reveal q exactly in the population, despite universal commitment and nonentry. Additional recorded details whose distribution is constant across q would not help.

Contests without commitment offer another route. Under the maintained response rules, only high types enter when $X = 0$, so the population entry rate there equals q . Different accounts therefore predict different entry rates. Because Incumbents choose commitment, however, committed and uncommitted contests may differ in Challenger populations or other determinants of entry. The researcher must justify carrying observations under $X = 0$ to the committed contests, through an intervention changing only commitment or a defensible comparison of populations and subsequent behavior.³ When that comparison is valid, the different entry rates satisfy Proposition 3(ii-b). A change that also alters payoffs, information or subsequent responses answers a different question.

Observing the Incumbent’s response to entry after commitment addresses threat credibility. An intervention inducing entry or evidence from comparable contests could establish whether the committed Incumbent fights. The meaning of that entry matters: an execution error, information about Challenger type, and a misspecified game can imply different responses (Appendix A.4). A corresponding intervention or comparability argument is required. By itself, this response does not reveal the proportion of high types and therefore does not narrow this probability of causation. The crisis-bargaining literature illustrates the empirical practice: response probabilities can be estimated at histories that repeated or comparable games reach and record, given an account of selection and why observations may be combined (Crisman-Cox and Gibilisco 2021).

Regime-specific outcome distributions suffice here because factual nonentry is certain. For other attribution questions they may leave the same-case pairing open. Evidence linking the same contests across regimes, when meaningful, or a defended structural restriction can constrain that connection; another separate sample from each regime generally cannot. Humphreys and Jacobs (2023) offer an approach to supplying this structure for case-level attribution. Within a fully specified causal model, they combine prior beliefs about a case’s responses with its observed evidence. The analysis here asks what the game and its solution determine about the missing response. Both approaches require its source to be stated: the observed outcome alone does not reveal how the same case would respond under another regime.

3. For naturally occurring uncommitted contests, a stability or ignorability condition requires entry under $X = 0$, conditional on the comparison variables, to have the same distribution there as it would among committed contests (Dawid 1979, 2021).

More observations improve precision when the relevant variation is present. Perfect separation in a finite sample can disappear as observations accumulate (Crisman-Cox, Gasparyan, and Signorino 2023). A history assigned probability zero by equilibrium never enters a sample from that equilibrium. Regularization may stabilize estimation by adding a restriction; it supplies no observations from that branch. The useful question is which disagreement new evidence resolves and which assumptions make it relevant to the target.

4.5 What Researchers Should Report

A report should let readers answer the paper’s three questions. The following seven checks connect the model’s quantities to the empirical record and make the sources of the answer explicit.

What is the model?

1. State the game form, information, payoffs, Nature’s probabilities, behavioral profile, solution concept and selection rule. Distinguish what the game specifies, what its solution implies and what is additionally assumed. Give the reasons or evidence for maintaining those claims, including claims about unobserved responses.

What is the causal question?

2. State the outcome, units or population, quantity sought, and regimes being compared; identify the factual regime when applicable.
3. Say what changes and remains fixed: a decision rule, an externally forced action, or the game itself. If the game changes, state whether the remaining rules are held fixed or selected anew.
4. Say whether the target uses only separate regime distributions or also depends on the same units’ outcomes across regimes. For the latter, state the connection or restrictions supplying the answer, or explain why the target is invariant to what remains open.

What is the evidence?

5. Specify the recording rule and observed regimes. Do the relevant histories occur with positive probability and enter the record? Which accounts giving different causal answers can the record distinguish? Justify comparisons across contests or populations.
6. Report whether the model, maintained assumptions and evidence leave one answer, informative bounds or the full range. State which ingredients support each part of the conclusion.
7. Name the measurement, variation, comparison or assumption that could narrow remaining disagreement. Additional observations from an unchanged distribution cannot reveal behavior it never generates.

Table 1 executes these checks for candidate entry.

5 Application: Candidate Entry

An incumbent deploys its campaign resources, and a challenger stays out. How much nonentry is attributable to deployment? The strategic politicians hypothesis holds that potential challengers weigh electoral prospects before entering (Jacobson and Kernell 1983); incumbents anticipate those choices. The numerical illustrations below are not election estimates. They distinguish what the game implies, what deployment, entry and entrant quality establish, and what additional evidence or assumptions would narrow attribution. The continuous type and deployment spaces make this an analogue of the finite-game analysis; the appendix derives its corner and identification results directly.

5.1 What Does the Entry Game Imply?

Resources deter entry in the model by raising the cost of challenging. Let $\bar{w}$ be a campaign budget acquired before the contest and $w \in [0, \bar{w}]$ the amount deployed. Acquisition cost is sunk, unused funds have no value after the contest, and deployment carries no additional marginal cost. This uses the terminal-spending logic of Goodliffe’s (2009) one-election model, while taking the available budget as given.

Each district has a potential challenger with type $V \sim F$ on a normalized $[0, 1]$ scale, where $F(v) = P(V \leq v)$. The incumbent knows F but chooses deployment without observing the realized type. The challenger observes its own type and the deployment before deciding whether to enter. Write $\pi_C(v)$ for the challenger's return from entering and $c(w)$ for the cost that deployment imposes on it. This cost function is distinct from the constant fight cost c in Section 2's entry game. The challenger enters exactly when

$$\pi_C(v) > c(w). \quad (3)$$

Assume both functions are strictly increasing and that their equality defines a unique cutpoint $v^*(w)$ at each available deployment. The cutpoint rises with deployment: types above it enter, and types at or below it stay out. Thus $Y_w = \mathbf{1}\{V \leq v^*(w)\}$ records nonentry under deployment w . The subscript identifies the deployment regime, written as a superscript in Section 3.1.

The incumbent values nonentry more than entry. Let π_I be its payoff against entry and $\Delta_I > 0$ the additional value of deterring it. Its expected payoff is

$$V_F(w) = \pi_I + \Delta_I F(v^*(w)),$$

where the subscript distinguishes expected payoff V_F from challenger type V . Assume F is absolutely continuous, with strict increase on $[v^*(0), v^*(\bar{w})]$, the interval between the lowest and highest available cutpoints. Then each increase in deployment deters some additional types, so the unique maximizing choice is full deployment, $w^* = \bar{w}$. I call this the regular distribution class. If flat portions of F are allowed, full deployment remains optimal but may require a selection rule when other deployments tie. The corner follows from these payoff and budget assumptions; a marginal deployment cost could instead produce an interior choice that reveals something about F . Appendix A.8.1 gives the derivation.

The factual regime uses full deployment. The alternative replaces the incumbent's decision rule with a fixed smaller deployment, $w_{\text{small}} < \bar{w}$, while holding the challenger's threshold rule and all other primitives fixed. This replaces the incumbent's strategy—a strategy replacement in Section 3.2's terminology—and the changed profile is not solved again. For the same-case comparison, declare one stable district type V under both regimes. This is the stable-type structural implementation: even if a tree displays separate Nature draws on different deployment branches, both branches use the same realized V . Together with the maintained threshold rule, V is a determining

concomitant in Definition 1’s sense. Its cross-regime stability is an assumption, not something the observed deployment establishes.

Figure 2 shows what attribution means under these rules. A challenger below both cutpoints would have stayed out under either deployment. A challenger between the cutpoints stays out under full deployment but would enter under the smaller one. Only the latter nonentry is attributed to the additional deployment. Knowing the actual challenger’s type would settle its classification under the model. Without that information, the available answer is a probability among comparable full-deployment nonentrants.

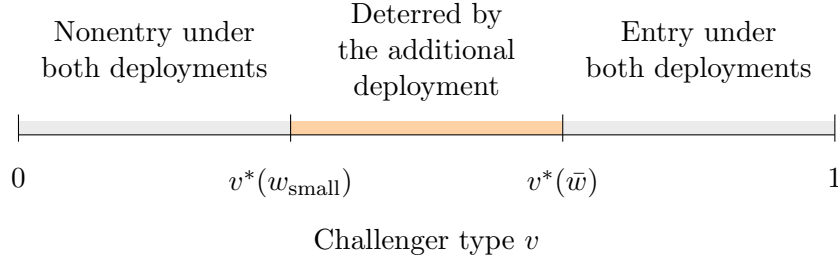


Figure 2: What deterrence means under the stable-type implementation. The same challenger type is used under both deployments. Full-deployment nonentrants lie at or below $v^*(\bar{w})$; those in $(v^*(w_{\text{small}}), v^*(\bar{w})]$ would enter under the smaller deployment. Equality at either cutpoint produces nonentry under that deployment. Positions are schematic; nonentrant types are not observed.

5.2 What Can the Election Record Establish?

The record can reveal the entry rate and the quality of entrants while leaving the composition of nonentrants unresolved. To isolate that limit, suppose even the population distribution of the recorded variables is known exactly. Use the numerical cutpoint $v^*(w) = 0.2 + 0.1w$, a common budget $\bar{w} = 4$, and the alternative $w_{\text{small}} = 1$. The two cutpoints are therefore 0.6 and 0.3. The researcher records deployment, entry, and the type of an entrant:

$$O_C = (w, \mathbf{1}\{\text{entry}\}, Q), \quad Q = V \text{ upon entry, } Q = \emptyset \text{ otherwise.}$$

The record contains neither nonentrant types nor identities of the unobserved type-specific loci in the game. Here $\emptyset$ records missing type information, not a zero-quality entrant.

Suppose every district deploys four units, 40% have entry, and entrant types are uniform on $(0.6, 1]$. The record fixes $F(0.6) = 0.6$ and the distribution above that cutoff. It therefore tells the researcher how much type probability lies below 0.6, but not how that mass is divided across 0.3. The incumbent's choice reveals only the binding budget under the maintained corner argument. It adds no further restriction on that division within the regular class.

Two complete accounts show the consequence. Under F_1 , the type masses below 0.3, between 0.3 and 0.6, and above 0.6 are respectively 0.55, 0.05, and 0.40. Under F_2 , the first two masses are exchanged. Both have the same 0.40 entry rate and entrant-type distribution and uniquely imply full deployment; both therefore reproduce the entire recorded law. Yet the deterred share among their nonentrants is $0.05/0.60 = 1/12$ in the first account and $0.55/0.60 = 11/12$ in the second. Panel (a) of Figure 3 displays these accounts. The disagreement concerns attribution despite identical observations, not the precision of an estimated entry rate.

Write $E = \{w = \bar{w}, Y_{\bar{w}} = 1\}$ for the factual full-deployment nonentry group. Its probability is $P(E) = F(0.6) = 0.6$. Under the stable-type implementation, the probability of causation is

$$\text{PC}_E = P(Y_{w_{\text{small}}} = 0 \mid E) = \frac{0.60 - F(0.30)}{0.60}.$$

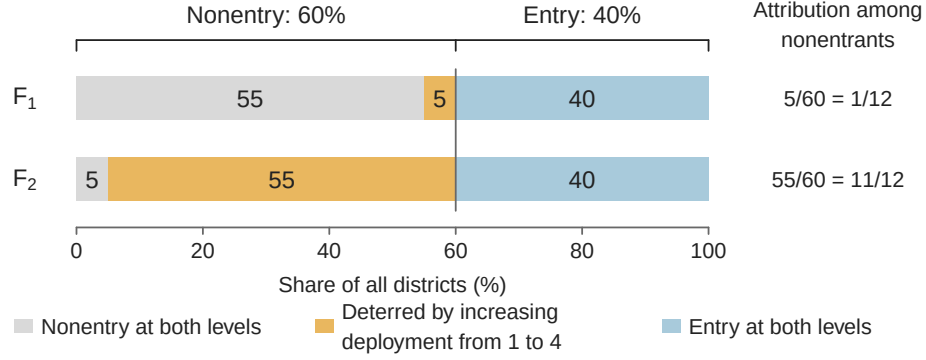
The numerator is the probability of a type between the two cutpoints; the denominator restricts the comparison to factual nonentrants. Unlike Section 4.3's example, the conditioning event is not certain. The same-case question is a goat before its cross-regime connection is supplied. Here the declared stable type supplies that connection, and the maintained threshold response makes nonentry monotone in deployment, reducing the target to the displayed ratio of type probabilities within this class. An equal type distribution under each regime, without that connection or another sufficient restriction, would not license the reduction.

Let $\mathcal{M}_C$ contain complete accounts $M(F, s)$, with all other primitives and the stable-type implementation fixed, where s selects a maximizing deployment. Let $\mathcal{M}_C^\circ$ retain the accounts reproducing the full O_C law. The answer set is

$$\mathbb{I}_{\text{PC}_E} = \left\{ \frac{0.60 - F(0.30)}{0.60} : M(F, s) \in \mathcal{M}_C^\circ \right\}. \quad (4)$$

Selection belongs in the account because a distribution permitting several

(a) Two accounts of the record at deployment 4



(b) Predictions that additional evidence could distinguish

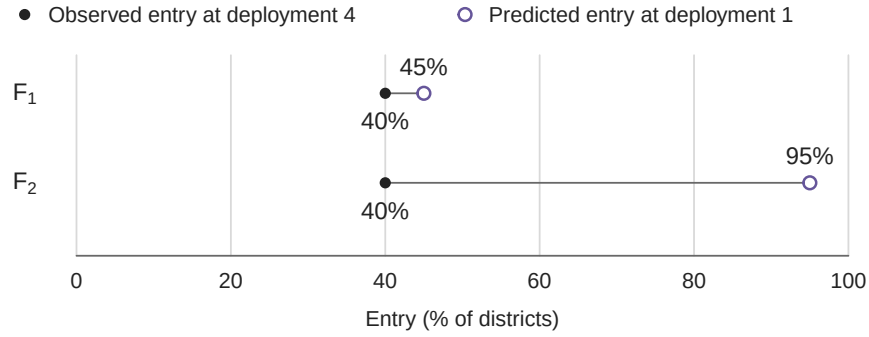


Figure 3: Same observations, different attribution. Both accounts match 60% nonentry, 40% entry, and the entrant-type distribution at deployment four. Panel (a)’s subdivisions are model-implied responses under the stable-type implementation, not measured nonentrant types. Among nonentrants, the fraction deterred by increasing deployment from one to four is $1/12$ under F_1 and $11/12$ under F_2 . Panel (b) shows predicted entry at deployment one, not observed entry under an actual change in resources.

optimal deployments need not generate the observed deployment under every selection. Appendix A.8.2 derives this application of Lemma 7 and proves sharpness.

On the regular class, the answer set is exactly $(0, 1)$. Strict increase requires positive type mass both below 0.3 and between 0.3 and 0.6, excluding both endpoints but allowing every interior value. On the relaxed class of absolutely continuous distributions allowing flat portions, the upper endpoint

requires $F(0.3) = 0$. It can be attained with positive density throughout $(0.3, 0.6)$, leaving full deployment uniquely optimal.

The lower endpoint requires $F(0.3) = F(0.6) = 0.6$, making deployments from one to four equally good. It is included under a rule selecting the upper corner or a class admitting all maximizing selections, and excluded under a rule that selects a lower maximizer whenever one exists. Thus the relaxed class gives $[0, 1]$ under the former treatments and $(0, 1]$ under the latter; requiring a unique maximizer also excludes zero. The endpoint distinction follows from maintained distribution and selection restrictions, not from an additional observation, paralleling the selection boundary in Corollary 1.

Even with entrant quality recorded and every nonentrant contest counted, equilibrium deployment leaves the alternative resource level unobserved. Repeating that record can improve finite-sample precision without narrowing the population answer set. The next question is which additional evidence could resolve the disagreement, and which evidence would instead support the assumptions used to translate the record into attribution.

5.3 Which Evidence Would Change the Answer?

Mechanism support and attribution evidence perform different jobs. For this threshold model, useful evidence connects deployable resources to entry costs, entry to costs and electoral prospects, and incumbent deployment to anticipated entry. Credible comparisons showing lower entry and, when entry occurs, higher average entrant type under greater deployment support that mechanism. They do not establish costless deployment, the exact threshold, or the missing nonentrant types.

Lee (2008) estimates party incumbency advantage using a close-election regression-discontinuity design. His finding that electoral defeat lowers a candidate’s probability of running again is consistent with entry responding to prospects. The treatment is barely obtaining incumbency, rather than a specified resource deployment. Applying this design to subsequent challenger entry and quality requires evidence that incumbency changes the relevant resources, that other consequences do not drive the comparison, and that the comparison can be carried to the target population. It supplies an effects-of-causes contrast, not automatically this attribution probability or the missing type distribution.

Caughey and Sekhon (2011) find pretreatment differences between bare winners and losers in close House races, challenging the RD comparison.

Strategic sorting can undermine that design without falsifying this game, which contains no vote-margin process; explaining the sorting requires an expanded game.⁴

Evidence narrowing attribution must restrict the portion of F that enters PC_E . Recording credible measures of potential-challenger type before the entry decision could do so even if every incumbent still deployed four units. Exact type measurement for nonentrants would classify them relative to 0.3 under the maintained threshold. Imperfect measurements or evidence from comparable populations could instead restrict $F(0.3)$ to an interval. For example, if independent information gives $F(0.3) \in [0.15, 0.45]$, the attribution set becomes $[1/4, 3/4]$ when that is the only additional restriction on the regular class. Those bounds come from the additional type information; the original record still supplies the denominator 0.6, and the declared stable type and maintained threshold response supply the translation. Appendix A.8.3 states the selection-invariance result for restrictions that keep attribution bounded above zero, including its precise scope.

Valid deployment variation offers another route. Under the maintained response rule, the population entry rate at deployment one is $1 - F(0.3)$. It would be 0.45 under F_1 and 0.95 under F_2 , so a valid comparison at that deployment could distinguish the two accounts. Panel (b) of Figure 3 contrasts these predictions with their common observed entry rate at deployment four. Seeing a lower deployment somewhere is insufficient: incumbent choices may reflect different challenger populations or different incentives. The comparison must justify carrying the type distribution and response rule to the factual full-deployment districts. Measuring a covariate or observing another contest helps only insofar as that argument makes it informative about the unresolved types.

Consider a separate uniform-type illustration. Budgets are assigned independently of challenger type in the target population; the maintained corner response converts assigned budgets into deployment. Assignment changes no other incentive or information, and districts do not interfere. Positive budget density around both comparison levels and continuity identify entry rates at exactly those levels. With V uniform on $[0, 1]$, the entry-rate curve is $0.8 - 0.1w$. The stable-type threshold rule supplies unit-level monotonic-

4. Signorino (1999)’s reanalysis of Bueno de Mesquita and Lalman (1992) shows why statistical models of strategic interaction should incorporate the theorized interdependence. That model-to-observation connection is distinct from asking which observations distinguish attribution accounts.

ity: additional deployment never induces entry. Together, the design and monotonicity identify attribution among full-deployment nonentrants as

$$\frac{F(v^*(4)) - F(v^*(1))}{F(v^*(4))} = \frac{0.6 - 0.3}{0.6} = 0.50.$$

This population result belongs to the uniform illustration; it is distinct from Figure 3’s two accounts and from the finite-sample diagnostic described in Appendix A.8.4. Independent assignment identifies the rates but does not establish monotonicity or their same-case connection. Without a sufficient restriction, known rates can leave attribution unresolved. In practice, shared party budgets could create interference, and resource shocks could change other incentives.

Mechanism evidence, type measurements and valid resource comparisons thus support different conclusions. Even an identified probability among comparable nonentrants need not classify one challenger’s decision; that requires enough case information to locate its response type.

5.4 Executing the Protocol

Table 1 applies Section 4’s reporting protocol to attribution among full-deployment nonentrants, distinguishing what the model implies, what the record establishes, and what additional evidence could resolve. The exercise supplies a form for an empirical report, not new estimates of electoral deterrence.

Table 1: Executing the protocol: full-deployment nonentrants.

Question	Answer in this application
What is the model?	
1. What features of the game and behavior are being maintained?	The game specifies actors, timing, information, payoffs, costs, and the sunk budget. Nature’s type law ranges over the declared class. Optimization implies full deployment on the regular class; stable type and tie-breaking are maintained. Credible comparisons showing lower entry and higher entrant type under greater deployment support this mechanism (Section 5.3) without locating types below 0.6. The threshold determines entry at both deployments.
What is the causal question?	

Continued on the next page

Table 1 (continued)

Question	Answer in this application
2. What is the causal question?	Among districts with nonentry under factual deployment four, what proportion would enter under deployment one? The outcome is nonentry, the population is E , and the quantity is the conditional probability PC_E .
3. What changes between those regimes, and what remains fixed?	Replace the incumbent's decision rule with deployment one. Hold the game, threshold response, and district type fixed. Challenger entry remains endogenous; there is no re-resolution.
4. Is the question a sheep or a goat?	The unrestricted question depends on how the same districts would fare under deployment one, not only on each regime's entry rate: a goat. The declared stable-type implementation supplies the connection; V determines both responses. Its monotonic threshold response permits the displayed ratio of type probabilities.
What is the evidence?	
5. What was actually observed, and what can those observations distinguish?	The record contains deployment, entry, and entrant type, including every nonentry contest. Only deployment four occurs, by the corner result; type is missing upon nonentry. F_1 and F_2 agree on the full record but imply 1/12 and 11/12. Their entry rates at deployment one differ; using such evidence requires a valid design or comparability argument.
6. How far does the answer go?	The record pins $F(0.6)$ and the entrant tail. Stable type translates remaining uncertainty into attribution. Strict increase gives $(0, 1)$; allowing flat distributions gives $[0, 1]$ when selection admits the zero endpoint, and $(0, 1]$ when it excludes it. The individual challenger's type remains unobserved.
7. What could narrow the answer?	Measure nonentrant types, defend restrictions on $F(0.3)$, or obtain valid deployment variation. The uniform-type comparison identifies 0.50 under its maintained assumptions, including monotonicity. Replicating the equilibrium record improves precision, not the population answer set.

A report can defend resource deterrence while leaving attribution anywhere in $(0, 1)$ on the regular class. Any narrower answer should identify the distinguishing evidence and maintained assumptions. This is a substantive mechanism claim with a stated evidentiary limit.

6 Conclusion

Representing an extensive-form game in Dawid’s decision-theoretic framework makes explicit what a causal claim about strategic behavior requires. The game supplies actors, possible moves, information, chance structure, and terminal consequences; behavioral rules determine how play proceeds, and a solution concept restricts those rules. These ingredients generate distributions under specified regimes, not evidence that the model describes the world. Some causal questions use only those separate distributions; others also depend on a connection between the same units’ outcomes across regimes. The latter require the source of that connection to be stated rather than inferred from the game’s solution.

Strategic behavior also shapes the evidence with which a causal claim can be assessed. Successful deterrence can suppress the behavior needed to learn its effect. Under the conditions of the deterrence result, accounts with different attribution answers produce the same recorded play even when behavior is uniquely selected. Exact knowledge of that recorded distribution does not separate them, and repeated observations from it cannot resolve their disagreement while the maintained assumptions and recording rule stay fixed. The candidate-entry application shows the related problem when every entry decision is recorded but nonentrant types and behavior under the alternative deployment remain unobserved.

Mechanism-based explanation remains a substantive achievement. A game can show how incentives, information, and expectations generate a pattern of behavior. Credible comparisons and case evidence may support the claim that this mechanism operates in the application without establishing what caused a particular outcome. The application also demonstrates how further measurements or a valid resource comparison can narrow attribution, provided the assumptions connecting that evidence to the causal question are defended. Independently assigned resources can identify regime-specific entry rates under suitable design assumptions; the further monotonicity restriction, not assignment alone, supports the attribution calculation. Supporting a mechanism and narrowing attribution are distinct achievements.

The reporting protocol makes these distinctions usable as a standard. A reader should be able to identify the causal comparison, see which parts of the answer follow from the game and behavioral assumptions, and understand what the observations distinguish. Where the answer remains open, the report should say which additional measurement, comparison, or restriction

would narrow it. An exact answer, informative bounds, or the full range can each be defensible under the evidence and maintained assumptions. The standard does not require every analysis to produce certainty. It requires enough information for readers to assess both the mechanism proposed and the grounds for carrying the causal claim to the world.

Data Availability Statement

Replication code accompanies the submission and will be deposited in the APSR Dataverse upon conditional acceptance. It reproduces Figure 3 from the two accounts' type masses and runs the separate uniform-type simulation described in Appendix A.8.4 as a finite-sample diagnostic. The article uses no external empirical data.

AI Use Disclosure

During revision, the author used Anthropic Claude (Table 5 and 5.1, via Claude Code) and OpenAI ChatGPT/Codex (GPT-5 and GPT-6 Astra), August–September 2026, to assist in analyzing the editorial and referee reports, organizing revisions, projecting word counts, and generating and editing draft prose. The author independently evaluated all suggestions, checked cited sources, verified the formal arguments and calculations, revised the resulting text, and assumes full responsibility for the manuscript.

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Online Appendix

Not intended for print publication

A Proofs and Technical Material

A.1 Structural representation

This subsection proves Proposition 1. The first two parts are direct correspondences; the work lies in the completeness claim in part (c).

Definition 2 (The augmented representation of a game). *Let Γ be a finite extensive-form game, and let $\emptyset$ be an indicator of nonreachability, not an action label. Let N be the set of players, and for each $i \in N$ let $\mathcal{H}_i$ be the collection of information sets belonging to player i . For each information set h , write $A(h)$ for its available actions and $\iota(h)$ for its owner, and introduce a typed decision variable D_h with pointed domain $A(h) \sqcup \{\emptyset\}$. For each Nature node t , write $A(t)$ for its available actions and introduce a typed chance variable W_t with pointed domain $A(t) \sqcup \{\emptyset\}$. These domains are pointed because $\emptyset$ is distinguished from the actions as their basepoint. The symbol $\sqcup$ denotes a disjoint union. The terminal coordinate Z takes terminal identities as its values.*

Let

$$V^{\text{mov}} = \{D_h : h \in \bigcup_{i \in N} \mathcal{H}_i\} \sqcup \{W_t : t \text{ is a Nature node}\}.$$

For a move variable v , let its locus $L(v)$ be the information set h when $v = D_h$ and the singleton node $\{t\}$ when $v = W_t$. A maximal root-to-terminal play p induces a structural assignment α_p : each move variable records the action taken if its locus is reached and $\emptyset$ otherwise, while Z records the terminal reached. This is well defined whenever no play visits an information set twice, a condition guaranteed by perfect recall, the standard requirement that players remember their own earlier actions and the information they possessed when taking them. Let

$$\mathcal{P}(\Gamma) = \{\alpha_p : p \text{ is a play of } \Gamma\}.$$

Let κ collect Nature's nodewise action probabilities, let $u = (u_i)_{i \in N}$ collect the players' payoff functions, and let kind record whether each variable is a

decision variable, a chance variable, or the terminal coordinate. The retained information is

$$\text{Inv}(\Gamma) = (V^{\text{mov}} \sqcup \{Z\}, \text{kind}, \iota, \text{pointed domains}, \mathcal{P}(\Gamma), \kappa, u).$$

An isomorphism between two such objects preserves every component, including player identities and the basepoint $\emptyset$. For a decision variable D_h , the set $R_h = \{\alpha \in \mathcal{P}(\Gamma) : \alpha(D_h) \neq \emptyset\}$ is its live event: the paths on which information set h is reached.

Let E_Γ be the set of directed edges obtained as follows. For distinct move variables v, w , include (v, w) in E_Γ when some node in $L(v)$ is a strict ancestor of some node in $L(w)$, and include (v, Z) for every move variable v . The augmented representation is $\text{Aug}(\Gamma) = (\text{Inv}(\Gamma), E_\Gamma)$. Call Γ layered when one linear order of its information sets and Nature nodes agrees with their order on every play.

Objects. Let $\mathfrak{G}$ be the class of finite, perfect-recall extensive-form games in which every information set and every Nature node has at least two actions. This is a structural-arity condition only: Nature's kernels may assign zero probability to an action. Perfect recall implies that no play visits one information set twice. Indeed, if a play reached h at a node y and again at a strict descendant y' , the player's experience at y' would contain an (h, a) pair absent from the experience at y , contradicting perfect recall. The structural assignment α_p in Definition 2 is therefore well defined.

Write $V^{\text{mov}} = V_D \sqcup V_W$ for the decision and Nature variables. For a structural assignment $\alpha \in \mathcal{P}(\Gamma)$, let

$$\text{supp}(\alpha) = \{v \in V^{\text{mov}} : \alpha(v) \neq \emptyset\}.$$

For any move variable v , write $R_v = \{\alpha \in \mathcal{P}(\Gamma) : \alpha(v) \neq \emptyset\}$. An isomorphism $\phi : \text{Inv}(\Gamma) \cong \text{Inv}(\Gamma')$ is a bijection of move variables and terminal identities, together with per-variable domain bijections, that preserves kind and ownership, fixes player indices and every basepoint $\emptyset$, maps $\mathcal{P}(\Gamma)$ assignment-wise onto $\mathcal{P}(\Gamma')$, carries each Nature kernel to its counterpart, and preserves payoffs componentwise.

Parts (a) and (b) now follow. There is one typed decision variable for each information set, carrying its owner, pointed action domain, and live event in the assignment set. Attaching a rule to that variable attaches one rule to the whole information set. Moreover, both a behavioral strategy and an

adapted regime are elements of $\prod_{h \in \mathcal{H}_i} \Delta(A(h))$, with $\delta_\emptyset$ off the corresponding live event. The map in Proposition 1(b) is thus the identity between two presentations of the same function space.

Relation to other transformation systems. Interchange also appears in broader transformation systems for games with the same reduced normal form (Thompson 1952; Elmes and Reny 1994). Those systems permit coalescing moves and adding superfluous moves, which change information and variable structure retained here and therefore answer a different equivalence question.

The relation. Suppose a decision node x belongs to information set h , and every child $x \cdot a$, $a \in A(h)$, belongs to one common information set k owned by a different player. An elementary interchange replaces the two layers

$$\{x\} \cup \{x \cdot a : a \in A(h)\}$$

by a k -node followed, after each $b \in A(k)$, by an h -node. The continuation subtree below action pair (a, b) is transplanted unchanged below pair (b, a) . The sets h and k retain their identities with their memberships transported node by node; every other information set, Nature node, and terminal is fixed. This surgery gives a canonical identification of the two invariants.

A labeled isomorphism is a node bijection preserving the rooted-tree order, deciders, information-set membership, player indices, action domains up to within-set renaming, Nature kernels, terminal identities, and payoffs. Let $\sim$ be the relation generated by elementary interchanges and labeled isomorphisms.

Legality, invariance, and inversion. An elementary interchange maps a game in $\mathfrak{G}$ to another game in $\mathfrak{G}$, preserves its invariant under the canonical identification, and is reversed by interchanging the new upper node with the common lower information set.

To see legality, first observe that no other node of h or k lies on a path through the displayed block: otherwise that play would visit the same information set twice. At the new k -node, its owner's experience is the same as at the old k -nodes because the transposed earlier move belongs to another player. At each new h -node, the owner of h has the same experience as at x .

For nodes in transplanted subtrees, each player's own sequence of information sets and actions is unchanged; the surgery merely transposes adjacent moves by different players. Perfect recall and structural arity are therefore preserved.

Plays through the block correspond by $(h = a, k = b) \mapsto (k = b, h = a)$; plays not through it are fixed. Corresponding plays induce the same assignments, reach events, and terminal identities. Nature nodes and their kernels are untouched, as are terminal payoffs. Hence the canonical map is an invariant isomorphism. The inverse block satisfies the same precondition and restores the original tree. It follows in particular that $\sim$ is an equivalence relation and that

$$\Gamma \sim \Gamma' \implies \text{Inv}(\Gamma) \cong \text{Inv}(\Gamma').$$

By definition, a labeled isomorphism also induces an invariant isomorphism, so the display covers both generators of $\sim$.

The converse uses five auxiliary facts.

Fan-out and terminality. If a node realizing a set of structural assignments $\mathcal{P}_x$ is nonterminal with move variable v , its action-child cells are nonempty, partition $\mathcal{P}_x$, and are all strict subsets. Finiteness ensures that every available action extends to a play, and structural arity supplies at least two nonempty cells. Consequently, the node is terminal exactly when $|\mathcal{P}_x| = 1$. At such a terminal, every variable in the support of the unique assignment has already been visited.

States and the placement of Nature. A *state* is an ordered tuple $x = ((v_1, a_1), \dots, (v_m, a_m))$ of distinct move variables with non-null values. Its cylinder is

$$\mathcal{P}_x = \{\alpha \in \mathcal{P}(\Gamma) : \alpha(v_j) = a_j \text{ for } j = 1, \dots, m\}.$$

A tree realizes x if one node's root path visits exactly those variables in that order with those values. That node is unique and its assignment set is $\mathcal{P}_x$. Terminality therefore transfers between two realizations of the same invariant that realize the same state.

Since W_t is unvisited at x , t cannot lie above the node at x . It is either that node, with $R_t = \mathcal{P}_x$; strictly below it in one action-child subtree, with

$R_t \subsetneq \mathcal{P}_x$ by fan-out; or elsewhere, with $R_t \cap \mathcal{P}_x = \emptyset$. Hence

$$\mathcal{P}_x \subseteq R_t \implies R_t = \mathcal{P}_x. \quad (\text{A.1})$$

Consequently, W_t is played at x in every realization of the invariant that realizes x . Thus two realizations cannot disagree at a shared state because one places a Nature move there. This is why Nature does not need, and is not allowed, to enter the interchange generator.

Admissibility. For a state x with $|\mathcal{P}_x| \geq 2$, call an unvisited move variable w *admissible at x* when:

- (i) $\mathcal{P}_x \subseteq R_w$;
- (ii) $\mathcal{P}_x \cap \{\alpha \in \mathcal{P}(\Gamma) : \alpha(w) = b\} \neq \emptyset$ for every $b \in A(w)$;
- (iii) if w is a Nature variable, $\mathcal{P}_x = R_w$; and
- (iv) if w is a decision variable, then for every unvisited Nature variable W_t , $\mathcal{P}_x \not\subseteq R_t$, and $R_t \subsetneq \mathcal{P}_x$ implies $R_t \subseteq \{\alpha \in \mathcal{P}(\Gamma) : \alpha(w) = b_t\}$ for some $b_t \in A(w)$.

These conditions depend only on $(\text{Inv}(\Gamma), x)$. The move actually played at a nonterminal node realizing x is admissible: (i)–(ii) follow from visitation and fan-out; (iii) follows from the Nature trichotomy; and, when w is a decision variable, (iv) follows because $\mathcal{P}_x \subseteq R_t$ would force W_t , rather than w , to be played at x by equation (A.1), while $R_t \subsetneq \mathcal{P}_x$ places that Nature node within one action-child subtree and hence one value of w .

Surfacing an admissible decision. Suppose a tree realizes x , an admissible decision variable $w = D_k$ is not the move currently played there, and the current node is nonterminal. A finite sequence of elementary interchanges confined to the subtree at x brings w to that node. Each interchange transposes, on every affected play, an adjacent pair (v, w) ordered v before w immediately beforehand, where v is unvisited at state x .

The proof is by strong induction on subtree height. The current move cannot be Nature by condition (iv) and equation (A.1); write it as D_h . The owners of h and k must differ. Because $\mathcal{P}_x \subseteq R_w$, every nonempty action-child cell reaches a k -node at or below that child. If one player owned both h and k , perfect recall would require all those k -nodes to have the same prior

own action at h , contradicting $|A(h)| \geq 2$. No child is terminal, because w is unvisited there but belongs to the support of every play through it.

The variable w remains admissible at each child state. Conditions (i)–(ii) follow from its occurrence and fan-out. For condition (iv), let W_t be unvisited at the child state. If $R_t \subsetneq \mathcal{P}_{x \cdot a}$, then $R_t \subsetneq \mathcal{P}_x$, so admissibility at x confines R_t to one value of w . If instead $\mathcal{P}_{x \cdot a} \subseteq R_t$, the Nature trichotomy at x forces $R_t = \mathcal{P}_{x \cdot a} \subsetneq \mathcal{P}_x$; admissibility at x again confines it to one value of w , contradicting fan-out for w within the child cell. By induction, surface w at every child where it is not already played. Each recursive interchange has the asserted property at that child state; a variable unvisited there is also unvisited at x . The child subtrees are disjoint, so those operations compose. Every child now belongs to information set k , and one final interchange brings w to x by transposing (D_h, w) , with D_h unvisited at x . Hence every step has the claimed property.

Reconstructing the tree from its paths. A realization is determined up to labeled isomorphism by the invariant and, for each structural assignment α , the visitation order π_α of $\text{supp}(\alpha)$. Map each node to its state. This map is injective by induction from the root and is onto the ordered prefixes of the schedules $\{\pi_\alpha\}$. The tree is therefore the trie of those prefixes: parenthood is prefix extension; the next variable fixes the decider and information set; action labels give extensions; Nature kernels attach to Nature variables; and complete prefixes carry terminal identities and payoffs. If two realizations carry literally the same invariant and schedules, this state-to-state isomorphism induces the identity on the common invariant.

Proof of Proposition 1(c). Soundness was established by legality and invariance above. For completeness, fix an isomorphism $\phi : \text{Inv}(\Gamma) \cong \text{Inv}(\Gamma')$. Relabel Γ along ϕ ; afterward the two trees realize one literal invariant. It is enough to connect these realizations by interchanges and then apply a labeled isomorphism inducing the identity on the common invariant. Write T for the relabeled tree and T' for Γ' ; the argument transforms T while holding T' fixed.

For a realization T , define

$$M(T) = \sum_{\alpha \in \mathcal{P}(\Gamma)} \# \{ \{v, w\} \subseteq \text{supp}(\alpha) : T \text{ and } T' \text{ order } v, w \text{ differently on } \alpha \}.$$

If $M(T) = 0$, all schedules agree and trie determination completes the proof. Otherwise choose a structural assignment whose two schedules differ. Because the schedules order the same support, their longest common prefix x , with the chosen assignment's values, is realized by both trees. They diverge before completion, so both nodes at x are nonterminal. Let w be the move played there in T' ; it is admissible at x in T' and hence, because admissibility depends only on the common invariant and x , in T . It cannot be Nature, since equation (A.1) would force T to play the same move at x . Surfacing therefore applies.

Each interchange used to surface w transposes, in the schedule of every affected assignment, an adjacent pair (v, w) ordered v before w immediately beforehand. Every affected assignment corresponds to a play through the node at x . In T' , w occurs at x , while v , which belongs to that assignment's support but is unvisited at x , occurs later. Each step therefore removes at least one disagreement and creates none, so $M(T)$ strictly decreases at every step. Iteration terminates because $M(T)$ is a nonnegative integer. Trie determination then supplies the final labeled isomorphism.

The construction first relabeled the game and then interchanged moves. Labeled isomorphisms preserve interchange sites and operations: the block precondition and transport use only tree order, deciders, membership, and action labels. Conjugating the interchanges by the initial relabeling therefore yields interchanges first and one labeled isomorphism last, with composite invariant map exactly ϕ . This proves the biconditional. The argument uses perfect recall in the once-per-path, legality, and different-owner steps, and structural arity in fan-out and surfacing; it never uses full support or layering. $\square$

The literal-edge interface. For the loci defined in Definition 2, a single elementary interchange with h above k satisfies

$$E_{\hat{\Gamma}} \triangle E_{\Gamma} \subseteq \{(D_h, D_k), (D_k, D_h)\}.$$

Relations to every third locus are unchanged: any node above the block remains above both layers, and every node in a transplanted subtree remains below both. Only the internal ancestry orientation can change. The statement is a containment, since an orientation created or removed at the local block may also be witnessed elsewhere by other nodes in the same information sets.

It follows that the raw literal graph is not a complete invariant. If augmented representations are identified exactly when their Inv components are isomorphic, then $\Gamma \mapsto [\text{Aug}(\Gamma)]$ induces the same quotient as $\Gamma \mapsto [\text{Inv}(\Gamma)]$. Reversed sequential encodings of a simultaneous game illustrate the point: their typed graphs have opposite orientations and are not player-fixed isomorphic, while one elementary interchange connects the underlying trees.

Finally, Γ is layered if and only if its literal move graph is acyclic. If it is layered, every edge respects the common linear order. Conversely, a topological order of an acyclic move graph respects every pair encountered in one play, because strict ancestry supplies the corresponding edge. Appending the terminal sink Z preserves acyclicity. This proves the acyclicity statement in Section 2.3.

Layering need not be preserved at each step of an interchange chain. To see this, let player 1 choose l or r ; on either branch, let player 2 move at an unobserving information set h , followed after every action by player 3 at an unobserving information set k . The original game is layered. Interchanging h and k only on the l branch creates both $D_h \rightarrow D_k$ and $D_k \rightarrow D_h$, so the intermediate game is not layered. Interchanging the r branch as well produces a layered endpoint again. Thus a chain between layered representatives may legitimately pass through the ambient nonlayered class.

Decisive boundary cases. Four cases fix the theorem's scope.

Reversed simultaneous encodings. The two trees share an invariant and are one interchange apart, while their literal typed edge orientations differ. This is the variation correctly removed by the quotient.

A unary move beside Nature. Let one player choose a or b , with b terminal; below a , place a one-action decision and a nondegenerate Nature move in either order, matching kernels, terminals, and payoffs. Their invariants are isomorphic: the decision and Nature variables have the same live event, and corresponding plays induce the same assignments. Yet they admit no interchange and are not labeled-isomorphic, witnessing failure of the biconditional without structural arity. Contracting the unary move changes the variable set and invariant.

An interlocked game. Player 1 chooses l or r at the root. After l , player 2 moves at node x_l ; after r , player 3 moves at node x_r . One action at x_l leads to a player-3 node and the other terminates; one action at x_r leads to a player-2 node and the other terminates. Let the player-2 nodes share

information set d_2 with actions a, b , and the player-3 nodes share d_3 with actions c, d . The six structural assignments, suppressing terminal identity, are

$$\begin{array}{lll} (l, a, c), & (l, a, d), & (l, b, \emptyset), \\ (r, a, c), & (r, b, c), & (r, \emptyset, d). \end{array}$$

Perfect recall and structural arity hold. No interchange site exists: the root has children in different information sets, each intermediate node has a terminal child, and the lower decision nodes have only terminal children. The literal graph contains $d_2 \rightarrow d_3$ and $d_3 \rightarrow d_2$, so the game is not layered while remaining inside the quotient theorem's domain.

Nondegenerate chance timing. If a two-action decision precedes an unobserved coin, the tree needs a separate Nature node below each decision action; if the coin precedes a pooled decision, it has one Nature node. The variable sets and hence the invariants differ. Both games satisfy structural arity, and the difference is retained because chance is never crossed.

Figure [A.1](#) records the principal graphical boundaries.

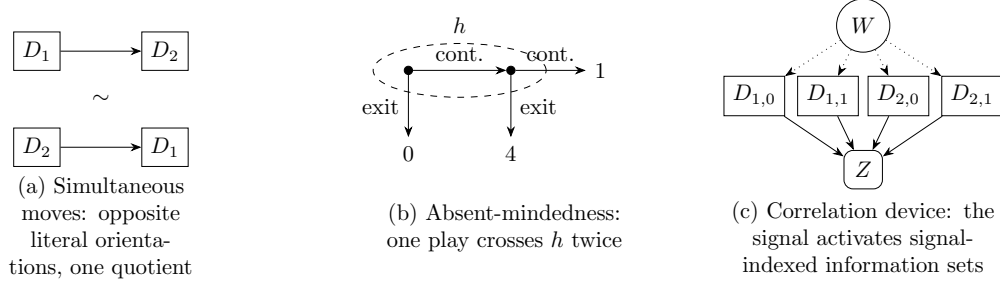


Figure A.1: Boundary cases for the structural mapping. Panel (a) shows two typed literal ancestry orientations for alternative sequential encodings of simultaneous play; descriptive interchange identifies their rich invariants rather than deleting both arrows. Panel (b) is excluded because the action at h is not one variable on a play that reaches h twice (Piccione and Rubinstein 1997). In panel (c), a parentless correlation device selects which signal-indexed information set is reached for each player; its arrows depict ancestry and activation, while the information sets encode which recommendation each player observes (Aumann 1974). Panel (c) omits the decision-to-decision ancestry arrows.

A.2 Profile-induced regime laws

Proof of Proposition 2. Fix a behavioral profile σ . Starting at the root, recursively assign to every partial history the product of the local kernels encountered along it. If a partial history reaches a player information set h , multiply by $\sigma_i(a \mid h)$ for each action-successor a ; if it reaches a Nature node t , multiply by $\kappa(a \mid t)$. At each node the successor probabilities sum to the probability already assigned to that node. Finiteness therefore yields a unique probability for every terminal history, and these probabilities sum to one. Mapping each terminal history to its structural assignment and terminal identity gives the unique law P_σ in part (a); pushing that law forward under ψ gives $P_\sigma(Y)$.

For part (b), an adapted regime g_i supplies one kernel in $\Delta(A(h))$ for every $h \in \mathcal{H}_i$, with $\delta_\emptyset$ off R_h . Its live-event kernels form a behavioral strategy regardless of structural arity, so (g_i, σ_{-i}) is another profile and the same recursion applies.

For part (c), compare (g_i, σ_{-i}) and (g'_i, σ_{-i}) . The intervention changes only the declared kernels at player i 's affected information sets. The nodewise

kernels κ of Nature, the behavioral kernels of every player $j \neq i$, and the terminal and outcome maps are identical under both regimes. This is equality of local kernels, not equality of the action or Nature draw realized in two different traversals. When Γ is layered, the literal augmented representation is a DAG and the recursively constructed law also admits the usual graph factorization; only in that case may the kernel-level restriction be restated as equality of the appropriate conditional distributions given graphical parents. $\square$

Kuhn's theorem can justify realization equivalence of mixed and behavioral strategies within one regime under perfect recall (Kuhn 1953). It does not select a joint seed law or common seed-to-action maps across two regimes.

A.3 Structural implementation

Definition 3 (Compatible structural implementation). *Fix a game, the rules σ_{-i}^* of the players other than i , and two regimes g_i, g'_i for player i . A structural implementation $\mathcal{I}$ consists of a common random element ξ taking values in $(\Xi, \mathcal{X})$ with law P_Ξ and, for every regime and every move locus, a deterministic measurable map from ξ to an action. A move locus is either a player's information set or a Nature node, and every map is defined even when that locus is not reached. A pure prescription is represented by a constant map. The implementation is compatible when traversal under each regime reproduces that regime's full profile-induced distribution of play and the two regimes' maps agree P_Ξ -almost surely at every locus whose rule is unchanged, including unchanged rules of player i . Maps may differ across regimes only at player i 's loci whose prescriptions change. Maps at distinct loci on regime-separated branches need not agree.*

Proof of Lemma 1. Fix a regime r . The recursive construction in the proof of Proposition 2 determines the probability of every terminal history and hence $\mu_r = P_{(r, \sigma_{-i}^*)}(Y)$. This construction is regime-specific and places no outcomes under different regimes on a common probability space, which proves part (a).

For part (c), consider the one-node binary game in the main text. With $U \sim \text{Unif}(0, 1)$, the nested maps

$$A^{g_i} = \mathbf{1}\{U \leq 1/4\}, \quad A^{g'_i} = \mathbf{1}\{U \leq 3/4\}$$

and the antithetic maps

$$A^{g_i} = \mathbf{1}\{U \leq 1/4\}, \quad A^{g'_i} = \mathbf{1}\{U > 1/4\}$$

are both compatible with the same two behavioral kernels and the same seed law. Yet the two actions disagree with probabilities $1/2$ and 1 . Their joint laws differ, proving that the kernels alone do not determine the cross-regime joint.

Now fix a compatible implementation $\mathcal{I}$. Given ξ and a regime r , its deterministic maps prescribe one action whenever traversal reaches a nonterminal node. Induction on depth therefore selects one successor at each reached node and, by finiteness, one terminal $z^{\mathcal{I}}(r; \sigma_{-i}^*, \xi)$. This terminal-history map is measurable. To see this explicitly, make terminal nodes absorbing. The root selector is measurable. If the current-node selector is measurable, partition Ξ into its finitely many node preimages and piece together the corresponding finite-valued measurable action and successor maps. The next-node selector is measurable; induction over the finite depth gives a measurable terminal selector. Composition with ψ makes $F_r^{\mathcal{I}}$ measurable.

The pair $(F_{g_i}^{\mathcal{I}}, F_{g'_i}^{\mathcal{I}})$ is therefore a measurable map on one probability space, and its pushforward of P_{Ξ} is the joint law displayed in Lemma 1(b). Compatibility gives each coordinate the marginal established in part (a). Conditional on the fixed implementation and the realized seed, both potential outcomes are constants. This proves part (b). $\square$

The binary example also exhibits the causes-of-effects consequence. If g'_i is observed and $A^{g'_i} = 1$, then

$$P(A^{g_i} = 0 \mid A^{g'_i} = 1) = \frac{2}{3}$$

under the nested implementation and equals 1 under the antithetic implementation. The omitted object is exactly the joint an attribution target can require.

Regime-separated randomness. Pure focal regimes do not eliminate the boundary. Suppose two regimes choose L and R deterministically. A fair binary Nature node W_L follows L and a distinct fair node W_R follows R , with Y equal to the realized bit. The nodewise kernels permit both $W_R = W_L$ and $W_R = 1 - W_L$. The potential outcomes are equal almost surely under the first implementation and differ almost surely under the second.

Nor does holding an opponent's behavioral profile fixed eliminate the problem. If an opponent randomizes fairly at distinct information sets h_L and h_R on the two regime branches, equal, opposite, and independent action

draws all preserve the opponent’s behavioral kernels while giving cross-regime disagreement probabilities 0, 1, and 1/2. Same-locus modularity does not determine a coupling across distinct loci.

A sufficient product-coordinate subclass. One convenient construction gives every locus ℓ a regime-specific uniform coordinate U_ℓ^r , imposes each regime’s within-regime product law, and lets the selector at ℓ depend only on U_ℓ^r through a map implementing that locus’s kernel. The regime-specific coordinate vectors may then be coupled, with action-map equality imposed at every unchanged same locus. Diagonal seed coordinates are sufficient but not necessary for that equality.

This product-coordinate/local-map construction is only a sufficient subclass of compatible master-seed implementations. It additionally imposes independence across mutually exclusive loci within a regime, whereas general compatibility constrains only collections of draws encountered on one realized path. Repackaging the coordinates of an unrestricted master seed as a nonproduct vector is a tautological recoding, not a converse characterization.

Two safe harbors. For the first, follow both regime traversals, realization by realization, to the first reached locus at which their prescriptions differ. Let H be the shared realized prefix ending immediately before that prescription is applied, with the natural terminal convention if no differing locus is reached. If all target-relevant randomness lies in H , the same realized H is used under both regimes, and every target-relevant continuation is deterministic conditional on (H, r) , then

$$Y^{g_i} = F_{g_i}(H), \quad Y^{g'_i} = F_{g'_i}(H).$$

The target-relevant joint is the pushforward of the shared-history law. This claim does not determine unused draws at mutually exclusive nodes.

For the second, the full joint may vary while a target τ is constant over every implementation compatible with the maintained restrictions. Every functional of the regime marginals has this property, and some causes-of-effects expressions acquire it after algebraic reduction or under additional restrictions. In that case implementation remains free for other questions but is not live for that target.

Compatible implementations are not recovered from the separate-regime play laws represented here: those laws constrain coordinate marginals, not their cross-regime coupling. Linked-unit or crossover evidence can change that information structure. When a target ranges over three or more regimes, the maintained class must contain one implementation jointly compatible with the whole regime family. Independently admissible pairwise couplings need not share a common joint extension. For example, each pair of three fair Bernoulli regime outcomes can separately be coupled to disagree almost surely, but no joint law can make all three pairwise disagreements hold simultaneously: three binary values cannot be pairwise unequal.

A.4 Three readings of an off-path entry

Suppose *Enter* occurs after the selected commitment, off the maintained equilibrium path. Three readings show why the antecedent alone does not fix a continuation. On an execution-tremble or forced-action reading, the maintained payoffs and strategic rules remain fixed. The Incumbent’s ordering therefore makes *Fight* the unique best response. Entry is implementation error, not evidence that the commitment lacks credibility.

On a payoff-inference reading, entry reveals a Challenger type θ richer than the maintained V . If θ enters only the Challenger’s payoffs, the Incumbent’s continuation ordering is unchanged. If it also changes the Incumbent’s expected fighting costs, however, the relevant comparison becomes

$$\mathbb{E}[u_I(\textit{Fight}, \theta) \mid \textit{Enter}] \geq \mathbb{E}[u_I(\textit{Accommodate}, \theta) \mid \textit{Enter}],$$

and either response or both may be optimal.

On a misspecified-form reading, the actual game permits the commitment to be withdrawn before the response. If withdrawal removes the accommodation cost, the uncommitted continuation ordering makes *Accommodate* optimal. That reading predicts entry followed by accommodation and removes the binding-commitment mechanism. The three readings vary execution, payoff information, and game form. Choosing among them requires an intervention account and any solution, refinement, and structural-implementation restrictions needed to define its admissible continuation; the words “suppose entry occurred” do not make that choice.

A.5 What observed play can distinguish

Formal objects. Fix a finite, perfect-recall representative game form $\mathcal{F}$, a solution-concept family $\mathcal{S}$, a recording map O , a target τ and its regime family D_τ , and all coordinate spaces. A complete account is

$$M = (u^M, \kappa^M, \sigma^M, \mathcal{I}^M),$$

where $\sigma^M \in \mathcal{S}(\mathcal{F}, u^M, \kappa^M)$ is the selected behavioral profile. Here $\mathcal{S}(\mathcal{F}, u^M, \kappa^M)$ is $\mathcal{S}(\Gamma)$ with the game's primitives written out. The coordinate u^M gives payoffs, κ^M gives Nature's nodewise kernels, and $\mathcal{I}^M$ is a compatible structural implementation. Let $\mathcal{M}$ be the set of complete accounts that satisfy the analyst's restrictions. When a selection rule is maintained, it is a separate selector χ satisfying $\sigma^M = \chi(u^M, \kappa^M)$; otherwise σ^M ranges over the permitted selections. The implementation $\mathcal{I}^M$ is required when the target, or another maintained object, uses a cross-regime joint distribution. For a marginal-only target, it may remain a nuisance coordinate without changing the target set. Nature's moves or randomized strategies alone do not require the analyst to fix an implementation.

For $r \in D_\tau$, let P_r^M be the full-play law supplied by Proposition 2 and Lemma 1. The recording map O is an analyst-specified measurable coarsening of completed play, and $P_{r,O}^M = O_\# P_r^M$. Write $P_O^M = P_{\sigma^M, O}^M$ for the population distribution of recorded play under account M 's selected profile. If P_O° is the population distribution actually observed, then

$$\mathcal{M}^\circ = \{M \in \mathcal{M} : P_O^M = P_O^\circ\}, \quad \mathbb{I}_\tau = \{\tau(M) : M \in \mathcal{M}^\circ\}.$$

Every conditioning event used by τ has positive probability under each $M \in \mathcal{M}^\circ$, unless a version is stipulated.

Proof of Proposition 3. Part (i). By definition, $\mathbb{I}_\tau = \{\tau(M) : M \in \mathcal{M}^\circ\}$. Because $\mathcal{M}^\circ$ is nonempty, this set is a singleton exactly when $\tau(M) = \tau(M')$ for every $M, M' \in \mathcal{M}^\circ$.

Part (ii-a). Once $(\mathcal{M}, O, P_O^\circ, \tau)$ is fixed, the equality defining $\mathcal{M}^\circ$ fixes that class and hence its target image. Repeated sampling through O from the same population distribution changes finite-sample precision, not that population set.

Part (ii-b). If $M, M' \in \mathcal{M}^\circ$, then $P_O^M = P_O^{M'} = P_O^\circ$. Thus every sample of the recorded variables under the two selected profiles has the same law. If their target values differ, both nevertheless belong to $\mathbb{I}_\tau$. If $P_{d,O}^M \neq P_{d,O}^{M'}$,

the population recorded law under d coincides with the prediction of at most one of the two accounts, so exact knowledge of that law excludes at least one of them. If instead $P_{d,O}^M = P_{d,O}^{M'}$, every sample recorded under d has the same distribution under both accounts, and no such observations separate them. $\square$

Lemma 2 (Determination within the maintained assumptions). *Assume $\mathcal{M}^\circ \neq \emptyset$. Fix the representative game form, payoffs, and Nature's kernels throughout the maintained class. If τ is constant over the entire jointly permitted set of remaining selected-profile and implementation pairs, then $\mathbb{I}_\tau$ is a singleton. Fixing the target-relevant profile and a compatible structural implementation wherever the target requires one is sufficient. When those objects are fixed, the maintained opponents' prescriptions together with the regime-invariant realization supplied by the implementation form a determining concomitant for a case-level target. Target invariance can establish the singleton conclusion without establishing one common concomitant across accounts. Observed play may be used to condition within an account, but it does not itself supply the implementation.*

Proof. Take $M, M' \in \mathcal{M}^\circ$. The joint-invariance hypothesis directly gives $\tau(M) = \tau(M')$. The nonempty target image is a singleton. Under the sufficient condition that the target-relevant profile and implementation are fixed, Lemma 1(b) supplies the common evaluation. Writing σ_{-i}^* for the fixed opponents' prescriptions, (σ_{-i}^*, ξ) is then regime invariant and determines the paired outcomes for a case-level target under the declared implementation. $\square$

The lemma's conclusion is conditional on the maintained class. A singleton does not establish that the fixed game form or behavioral assumptions describe the world correctly.

Definition 4 (Witness and exposure). *Let $\mathcal{H}$ be a declared finite family of information sets and let $R_{\mathcal{H}} = \bigcup_{h \in \mathcal{H}} R_h$ be the event that one of them is reached. Suppose every $M \in \mathcal{M}^\circ$ assigns $P_{\sigma_M}^M(R_{\mathcal{H}}) = 0$, and fix $d \in D_\tau$ such that $P_d^M(R_{\mathcal{H}}) > 0$ for every such M . An $(\mathcal{H}, d)$ -witness pair consists of $M, M' \in \mathcal{M}^\circ$ that give different target values and agree outside a predeclared $\mathcal{H}$ -gated coordinate block. The block is $\mathcal{H}$ -gated when every difference it induces under d , or in the target's evaluation, is confined to the distribution of first arrival over $\mathcal{H}$ and/or to primitives, prescriptions, and continuation*

play at or following $\mathcal{H}$, with target-relevant coordinates elsewhere fixed. At a null-arrival information set, this condition concerns maps, kernels, prescriptions, and continuation structure rather than a conditional distribution given arrival. The witness pair is $\mathcal{H}$ -exposed when $P_{d,O}^M \neq P_{d,O}^{M'}$.

Lemma 3 (What a witness establishes). *The two target values supplied by an $(\mathcal{H}, d)$ -witness pair belong to $\mathbb{I}_\tau$, and no sample recorded under the selected profiles distinguishes the pair. Population recorded play under d distinguishes the pair exactly when it is $\mathcal{H}$ -exposed.*

Proof. Membership of M and M' in $\mathcal{M}^\circ$ puts both unequal target values in $\mathbb{I}_\tau$ and gives both accounts the same recorded population law. The definition of exposure is exactly the condition that their recorded population laws under d differ. $\square$

Lemma 4 (Scope of sources of target variation). *(a) Within a subclass of $\mathcal{M}^\circ$ that fixes every coordinate outside a declared $\mathcal{H}$ -gated block, a singleton target image means that isolated variation in that block creates no variation in the answer within that subclass. It implies nothing about coordinates constrained to move jointly.*

(b) Within a nonempty subclass that fixes every nonimplementation coordinate, including every represented regime-specific play law, the target image over compatible structural implementations is a singleton exactly when the target is invariant across those implementations. Separate-regime evidence cannot expose residual implementation variation; maintained structure or genuinely linked evidence is required. This residual is not variation caused by failure to visit an information set.

(c) Within a nonempty subclass that fixes every nonselection coordinate, the target image over permitted selected profiles is a singleton exactly when the target is invariant across those profiles.

None of these statements supplies an intrinsic or additive decomposition of the total variation in $\mathbb{I}_\tau$.

Proof. For part (a), restrict $\mathcal{M}^\circ$ to the stated subclass. A singleton image is, by definition, absence of variation from isolated movement of the block within that subclass. This is neither a contrapositive nor a conclusion about coordinates constrained to move together. If the class permits no isolated variation in a block, the singleton condition holds vacuously.

For part (b), restrict instead to a subclass fixing every regime-specific play law represented by the separate-regime evidence. The compatibility requirement in Section 3.3 makes variation among compatible implementations invisible to every such law and hence to every recorded coarsening. Lemma 1(c) establishes that distinct compatible implementations can nevertheless remain. The target image is a singleton exactly when τ is invariant across them. Linked-unit or crossover evidence changes the evidence structure rather than supplying another separate-regime marginal.

For part (c), fix all nonselection coordinates. The set of possible target values is precisely the image of the permitted selected profiles, which is a singleton exactly when the target is constant across them. $\square$

Lemma 5 (How additional information can narrow the target set). *For a maintained subclass $\mathcal{M}_j \subseteq \mathcal{M}$, write*

$$\mathbb{I}_\tau(\mathcal{M}_j) = \{\tau(M) : M \in \mathcal{M}_j, P_O^M = P_O^\circ\}.$$

(a) *For fixed (O, P_O°, τ) , if $\mathcal{M}_1 \subseteq \mathcal{M}_2$, then*

$$\mathbb{I}_\tau(\mathcal{M}_1) \subseteq \mathbb{I}_\tau(\mathcal{M}_2).$$

(b) *Suppose that, for a measurable map h , $O_0 = h \circ O_1$ and $P_{O_0}^\circ = h_\# P_{O_1}^\circ$. Consistency with the richer recording O_1 then implies consistency with O_0 , so the richer recording weakly shrinks the target set.*

(c) *Adding a validly transported recorded-data law from another regime conjoins another restriction and therefore weakly shrinks the target set.*

Proof. Part (a) follows because intersecting a smaller maintained class with the equality defining observational consistency yields a subset of the class produced from the larger maintained class. Part (b) follows by applying h to the equality of the distributions under O_1 . Part (c) follows because an additional valid equality further restricts the accounts that survive. $\square$

For an ordered scalar target, point identification, informative bounds, and the full maintained range describe possible positions of $\mathbb{I}_\tau$. They are not another classification of what the causal question depends on.

A witness template. Starting from $M \in \mathcal{M}^\circ$, a perturbation M' supplies an $(\mathcal{H}, d)$ -witness when the following checks succeed:

- (t0) $M' \in \mathcal{M}$: it satisfies every maintained coordinate restriction and structural-implementation compatibility condition.
- (t1) The complete solution conditions, including belief consistency, hold at M' .
- (t2) Selection does not migrate outside the block. If χ is maintained, recompute $\sigma^{M'} = \chi(u^{M'}, \kappa^{M'})$ and verify the required agreement; otherwise verify it directly for the selected-profile coordinates.
- (t3) The observational recorded-data law, not necessarily the latent full-play law, is unchanged:
$$P_O^{M'} = P_O^M = P_O^\circ.$$
- (t4) Every induced difference, including any allowed selection change, is structurally $\mathcal{H}$ -gated.
- (t5) $\tau(M') \neq \tau(M)$.
- (t6) If exposure is claimed, $P_{d,O}^{M'} \neq P_{d,O}^M$.

Conditions (t4) and (t6) do different work: gating locates the difference, while exposure asks whether the proposed additional evidence can reveal it.

Payoffs at terminals reachable only through $R_{\mathcal{H}}$ can form the varied block if (t0)–(t5), especially the full solution conditions and selector stability, are separately verified. Nature kernels at chance nodes reachable only through $R_{\mathcal{H}}$, or latent components whose target-relevant effect under d is confined to $R_{\mathcal{H}}$, can do the same after their incentive and selection effects are checked. Slack in one equilibrium inequality does not establish these conditions because a payoff-dependent selector can migrate even when every inequality of the original selected profile remains satisfied.

A.6 Concealment lemmas

Lemma 6 (Equilibrium concealment). *Let $X \in \{0, 1\}$ be preventive and $Y \in \{0, 1\}$, with $Y = 1$ protected. Suppose the maintained class is $\{M(\nu) : \nu \in \mathcal{N}\}$ and varies only the law ν of a latent V . Suppose also that maintained*

structure fixes $Y(1, v) = 1$ for every v and $Y(0, v) = 0$ exactly when $v \in B$, for a measurable set B . Assume every $M(\nu)$ induces the same recorded law, placing probability one on $(X, Y) = (1, 1)$, and that the population recorded-data law equals that common law. Then

$$\text{PC} = P(Y^{X=0} = 0 \mid X = 1, Y = 1) = \nu(B), \quad \mathbb{I}_{\text{PC}} = \{\nu(B) : \nu \in \mathcal{N}\}.$$

Because the conditioning event has probability one, no cross-regime pairing affects this target. Additional draws from the same recorded law leave the set unchanged.

Proof. Under every $M(\nu)$, the factual event $(X, Y) = (1, 1)$ has probability one. Conditioning on it is therefore vacuous. The maintained response relation under Proposition 2(b)'s deviation—which holds all other prescriptions fixed—gives $Y^{X=0} = 0$ exactly on $\{V \in B\}$, so $\text{PC} = \nu(B)$. By the common-law and population-law assumptions, every $M(\nu) \in \mathcal{M}$ is observationally consistent, so $\mathcal{M}^\circ = \mathcal{M}$. Taking its image gives $\mathbb{I}_{\text{PC}} = \{\nu(B) : \nu \in \mathcal{N}\}$. Because the population recorded-data law is common to the class, further samples from that law do not alter the set. The reduction is to a deviation-regime marginal, so compatible cross-regime couplings do not change this target. $\square$

Lemma 7 (Conditional concealment). *Let a declared structural implementation contain one regime-invariant type V , with potential outcomes deterministic functions $Y(1, V)$ and $Y(0, V)$. Let E be determined by the recorded observables, with $P^M(E) > 0$ for every $M \in \mathcal{M}^\circ$, and suppose $(X, Y) = (1, 1)$ throughout E . Assume the entire recorded-data law, on and off E , is common to the admissible class and equals the population law, and that $Y(0, v) = 0$ exactly when $v \in B$ on the support of $V \mid E$. Then, writing ν_E^M for that conditional type law,*

$$\text{PC}_E^M = P^M(Y^{X=0} = 0 \mid E) = \nu_E^M(B), \quad \mathbb{I}_{\text{PC}_E} = \{\nu_E^M(B) : M \in \mathcal{M}^\circ\}.$$

The set is invariant to replication from the same recorded-data law.

Proof. On E , factual protection is certain. The declared implementation uses the same V under the deviation, and the deterministic response relation makes the counterfactual failure event $\{V \in B\}$. Conditioning therefore gives $\nu_E^M(B)$ for each M . The displayed target image and replication claim follow from the definition of $\mathcal{M}^\circ$. $\square$

The regime-invariant type is load-bearing. If the factual and counterfactual regimes instead draw independent types with the same marginal F , factual conditioning does not select the counterfactual type. This preserves each regime marginal while changing their coupling. For the pair used in Section 5, at that section’s counterfactual cutpoint $v^*(w_{\text{small}}) = 0.3$, the target becomes $1 - F(0.3)$, yielding 0.45 and 0.95 rather than the stable-type values. The marginal threshold rules have not failed; the implementation has changed.

A.7 Deterrence: derivation and bounds

Normalize the Incumbent’s payoff from uncontested office to zero. Let $\pi_I < 0$ be its payoff from accommodating entry, let $-c_F$ be its payoff from fighting, and let $k > 0$ be the additional cost of backing down after commitment. Commitment has no direct payoff consequence when entry does not occur. The ordering $\pi_I > -c_F > \pi_I - k$ makes the Incumbent accommodate after uncommitted entry and fight after committed entry. Both Challenger types stay out against a fight; against accommodation, H enters and L stays out. Under these maintained payoff orderings, sequential rationality pins every continuation prescription. Because both players observe (X, V) , the relevant information sets are singletons. Writing $q = P(V = H)$, commitment is strict for $q > 0$ and tied with noncommitment at $q = 0$. At that endpoint, the root prescription depends on the maintained selection treatment. The deviation changes only the commitment-node prescription to $X = 0$ and holds every continuation prescription fixed.

The recording map is $O = (X, Y)$; it records neither V nor type-indexed locus identity. Under a committing selection its law is a point mass at $(1, 1)$ for every q , although the latent full-play law varies with the type frequency. A richer recording that logs type-indexed locus identity would reveal V through the visitation pattern and break the common-law claim; the recording map is therefore part of the result. With $B = \{H\}$, Lemma 6 gives

$$\text{PC} = P(Y^{X=0} = 0 \mid X = 1, Y = 1) = P(Y^{X=0} = 0) = q.$$

Dropping the conditioning is a probability-one marginal reduction; only after that reduction is cross-world coupling irrelevant to this PC.

The endpoint depends on the maintained selection class. Selecting commitment at indifference admits $q = 0$ and gives $[0, 1]$. A uniquely selecting

rule that chooses noncommitment at $q = 0$ excludes that specification from the observed committing law and gives $(0, 1]$. A class admitting all solution-consistent selections includes the committing selection at $q = 0$ and restores $[0, 1]$. If $q \in [a, b]$, the first and third classes give $[a, b]$, as do all three when $a > 0$; the second gives $(0, b]$ when $a = 0$. These sets maintain the deviation-branch payoff orderings. If those orderings also vary, the possible response sets include $B = \emptyset$, $B = \{L\}$, and $B = \{L, H\}$, adding 0, $1 - q$, and 1; full ignorance still unions to $[0, 1]$. When $a = b = 0$ under the noncommitment unique-selector branch, $\mathcal{M}^\circ = \emptyset$: that maintained selector is inconsistent with the observed committing law, so no identified set is reported for that frame.

For the witness classification, let $\mathcal{H}$ be the family consisting of the two type-indexed Challenger information sets following $X = 0$, and let d impose the $X = 0$ prescription. Under every admissible observationally consistent committing specification, $P_{\sigma^M}^M(R_{\mathcal{H}}) = 0$, whereas $P_d^M(R_{\mathcal{H}}) = 1$. For any two such specifications with $q \neq q'$, the observational recorded-data laws agree. Under d , changing q changes only the first-arrival distribution over $\mathcal{H}$, namely $(1 - q, q)$, while the continuation prescriptions remain structurally fixed. The block $\{q\}$ is therefore $\mathcal{H}$ -gated and the pair $\mathcal{H}$ -exposed: the recorded deviation-regime entry rate equals q . Stable-type coupling is instead residual implementation variation, never separated by these regime-specific laws; it is not live for this PC because of the marginal reduction.

For completeness, evaluate the combined-data bounds of Tian and Pearl (2000, eq. (25)), whose lower bound originates with Robins and Greenland (1989). Here x and x' denote $X = 1$ and $X = 0$, y and y' denote protection and failure, and subscripts denote interventions:

$$\max \left\{ 0, \frac{P(y) - P(y_{x'})}{P(x, y)} \right\} \leq \text{PC} \leq \min \left\{ 1, \frac{P(y'_{x'}) - P(x', y')}{P(x, y)} \right\}.$$

For a fixed q , $P(y) = P(x, y) = 1$, $P(x', y') = 0$, $P(y_{x'}) = 1 - q$, and $P(y'_{x'}) = q$, so both bounds equal q . Equilibrium observations do not supply the interventional marginal; taking the union over admissible q yields the corollary's set. Independently known components compute it, observations under $X = 0$ can expose it subject to design and transport, and additional observations inside an already supported comparison improve precision. These are population bounds for recorded peaces. Case attribution conditions further on the case's observables and, when its type remains latent, stops at the posterior for that covariate cell (Dawid, Humphreys, and Musio 2024).

A.8 Candidate-entry derivations

A.8.1 The deployment corner

Write $\mathfrak{F}$ for the regular class of absolutely continuous type distributions strictly increasing on $[v^*(0), v^*(\bar{w})]$, and $\overline{\mathfrak{F}}$ for the relaxed class of absolutely continuous distributions allowing flat portions. In the numerical application both classes are supported on $[0, 1]$.

Let $w \in [0, \bar{w}]$, let $\Delta_I > 0$ be the incumbent's value of deterring entry, and write

$$V_F(w) = \pi_I + \Delta_I F(v^*(w)).$$

Here V_F denotes incumbent value; V without the subscript is the stable challenger type. For $w_2 > w_1$, strict increase of v^* and of F over the relevant cutpoints gives

$$V_F(w_2) - V_F(w_1) = \Delta_I [F(v^*(w_2)) - F(v^*(w_1))] > 0.$$

Thus $w^* = \bar{w}$ uniquely on the regular class $\mathfrak{F}$. If F is absolutely continuous with density f , strict increase on $[a, b]$ is equivalent to

$$\int_c^d f(v) dv > 0 \quad \text{for every } a \leq c < d \leq b.$$

The familiar condition $f > 0$ almost everywhere is sufficient but strictly stronger. The relaxed class $\overline{\mathfrak{F}}$ contains absolutely continuous distributions with $f \geq 0$ almost everywhere. On that class V_F is weakly increasing and $\bar{w}$ remains a maximizer; the upper-corner convention selects it when the maximizer is not unique.

This conclusion uses the terminal model's costless deployment of a previously acquired budget. If deployment instead bears an unsunk differentiable cost $C(w)$, an interior optimum can satisfy

$$\Delta_I f(v^*(w)) v^{*'}(w) = C'(w),$$

so the observed choice may reveal local information about F .

A.8.2 The sharp attribution set

The recording map is

$$O_C = (w, \mathbf{1}\{\text{entry}\}, Q), \quad Q = \begin{cases} V, & \text{upon entry,} \\ \emptyset, & \text{otherwise.} \end{cases}$$

The literal missing-value symbol $\emptyset$ keeps nonentry distinct from entry by a zero-quality challenger. Fix either $\mathfrak{F}$ or $\overline{\mathfrak{F}}$. Let $\mathcal{M}_C$ contain the complete accounts $M(F, s)$ for which F belongs to that class and s is an admissible selection from the maximizing-deployment correspondence. Define

$$\mathcal{M}_C^\circ = \{M(F, s) \in \mathcal{M}_C : P_{s, O_C}^{M(F, s)} = P_{O_C}^\circ\}.$$

Here the subscript s stands for the profile it selects. Apply Lemma 7 with admissible class $\mathcal{M}_C^\circ$; its observationally consistent subclass is then itself. The defining equality supplies the common entire recorded-data law required by the lemma. The application assumes that $P_{O_C}^\circ$ is generated inside $\mathcal{M}_C$, so $\mathcal{M}_C^\circ$ is nonempty. Membership jointly matches deployment, the entry rate, and the entrant-quality law. Because O_C records deployment, consistency retains only selections that choose the observed $\bar{w}$. The corner result makes this automatic in $\mathfrak{F}$; in the relaxed class the selection coordinate performs the needed filtering. Omitting s would lose this restriction when deployment is nonunique.

Code $X = 1$ as the factual $\bar{w}$ -deployment regime, code $X = 0$ as the deviation to w_{small} , and code $Y = 1$ as nonentry. With the numerical illustration's parameters, $\bar{w} = 4$, $w_{\text{small}} = 1$, $v^*(\bar{w}) = v^*(4) = 0.6$, and $v^*(w_{\text{small}}) = v^*(1) = 0.3$; the recorded nonentry rate is 0.6. Let

$$E = \{w = \bar{w}, \text{ nonentry}\}, \quad P^M(E) = F(0.6) = 0.6.$$

The equality holds for every $M = M(F, s) \in \mathcal{M}_C^\circ$. For each such M , let $\nu_E^M = \mathcal{L}_M(V \mid E)$, the normalized restriction of F to $V \leq 0.6$. The declared implementation uses the same V under both deployments. With the lemma's B equal to the attribution interval $(0.3, 0.6]$,

$$Y(0, v) = \mathbf{1}\{v \leq 0.3\}, \quad Y(0, v) = 0 \iff v \in B$$

on $\text{supp}(V \mid E) \subseteq [0, 0.6]$. The lemma consequently gives

$$\text{PC}_E^M = \nu_E^M(B) = \frac{F(0.6) - F(0.3)}{F(0.6)} = \frac{0.6 - F(0.3)}{0.6},$$

and its image over $\mathcal{M}_C^\circ$ is equation (4).

A single family establishes sharpness. For $p \in [0, 1]$, let G_p be the absolutely continuous distribution on $[0, 1]$ with density

$$g_p(v) = \begin{cases} 2(1-p), & 0 < v < 0.3, \\ 2p, & 0.3 < v < 0.6, \\ 1, & 0.6 < v < 1. \end{cases}$$

Every member has $G_p(0.6) = 0.6$ and the same observed entrant tail: its unconditional density is 1 above 0.6, hence its conditional density among entrants is 2.5. For any p admitted by the chosen class, an admissible selection s that chooses the observed $\bar{w}$ therefore gives $M(G_p, s) \in \mathcal{M}_C^\circ$: deployment, the entry rate, and the entrant-quality law all match. Moreover,

$$G_p(0.3) = 0.6(1 - p), \quad \text{PC}_E^{M(G_p, s)} = \frac{0.6 - G_p(0.3)}{0.6} = p.$$

The manuscript's two witnesses are $F_1 = G_{1/12}$ and $F_2 = G_{11/12}$.

For $0 < p < 1$, the density puts positive mass on every subinterval of the relevant cutpoint range, so every value in $(0, 1)$ is attained in the regular class. Conversely, strict increase for every regular-class member gives

$$0 < F(0.3) < F(0.6) = 0.6,$$

where the first inequality follows from $F(0.3) > F(0.2) \geq 0$. Hence $0 < \text{PC}_E^M < 1$. Thus the regular-class image is exactly $(0, 1)$. At $p = 1$, $G_1(0.3) = 0$ and g_1 remains positive throughout $(0.3, 0.6)$; $\bar{w}$ is still the unique maximizer, so relaxation alone admits the upper endpoint. At $p = 0$, $G_0(0.3) = G_0(0.6) = 0.6$, making incumbent value flat over deployments whose cutpoints span $[0.3, 0.6]$. The lower endpoint survives under the upper-corner convention or a class admitting every maximizing selection, but not under a maintained lower selection or a uniqueness requirement. Hence $[0, 1]$ is sharp under relaxation with an endpoint-including selection treatment.

If the two regimes instead draw independent types with the same marginal F , conditioning on factual nonentry does not select the counterfactual type. The target then becomes $1 - F(0.3)$, equal to 0.45 for F_1 and 0.95 for F_2 . Those are different targets under a different implementation, not approximations to $1/12$ and $11/12$.

A.8.3 Selection and externally restricted answer sets

The selection comparison holds the numerical cutpoints, recorded law, and stable-type implementation fixed. A selection treatment assigns to each distribution a nonempty collection of maximizing-deployment selections; a fixed maximizing rule and the collection of all maximizing selections are examples. No additional distribution restriction is imposed in this comparison. An alternative treatment may require a unique maximizer, retaining every distribution that has one. Denote the resulting target image by J .

On $\mathfrak{F}$, $J = (0, 1)$ under all these treatments. On $\overline{\mathfrak{F}}$, every treatment admits all of $(0, 1]$. It also admits zero if and only if at least one distribution with $F(0.3) = F(0.6) = 0.6$ and the recorded entrant tail is admitted together with a selection choosing the observed deployment four with probability one. Thus $J = [0, 1]$ under the upper-corner treatment or all maximizing selections, and $J = (0, 1]$ under a rule always choosing a strictly lower maximizer when one exists or under a unique-maximizer requirement.

To prove the positive-value claim, take the family G_p above. For every $p \in (0, 1]$, its density is strictly positive on $(0.3, 0.6)$. For any $w < 4$, the interval between $v^*(w)$ and 0.6 consequently has positive mass, so deployment four is the unique maximizer. Every maximizing selection chooses it, its recorded law matches the observed one, and its target equals p . For $0 < p < 1$, G_p is also regular. The previous endpoint argument proves necessity at zero and excludes both endpoints in the regular class. This establishes the claimed images.

Now add only the target restriction $\text{PC}_E \in [a, b]$, where $0 \leq a \leq b \leq 1$; equivalently, $F(0.3) \in [0.6(1 - b), 0.6(1 - a)]$. The new image is exactly $J \cap [a, b]$, since the restriction filters accounts only by their target value and the witnesses attain every value in J . In particular, on the relaxed class, if $a > 0$ the image is $[a, b]$ under every selection treatment just described. If $a = 0 < b$, it is $[0, b]$ or $(0, b]$ according to whether zero is admitted; if $a = b = 0$ and zero is excluded, the restricted class is inconsistent with the recorded law. On the regular class the image is $[a, b] \cap (0, 1)$. The example $F(0.3) \in [0.15, 0.45]$ therefore gives $[1/4, 3/4]$ on either distribution class under every such treatment. Restrictions on other features of F can remove the witnesses, so this claim does not cover arbitrary joint restrictions on the distribution.

This is invariance of the target image, not uniqueness for every account with a positive target. For example, a density equal to 1 on $(0, 0.3)$, 1.5 on $(0.3, 0.5)$, zero on $(0.5, 0.6)$, and 1 on $(0.6, 1)$ has the same recorded entrant tail and $\text{PC}_E = 1/2$. Its maximizing deployments are all of $[3, 4]$. A lower selection excludes that account from the observed-deployment class, but $G_{1/2}$ retains the same target value with a unique maximizer.

A.8.4 What the external resource comparison identifies

Let the externally assigned budget be independent of V and have positive density in neighborhoods of one and four. Maintain full deployment of that

budget, the common threshold rule, and no interference between districts. For the comparison to inform the original full-deployment population, maintain the same type law and response rule there; a different comparison population requires an additional transport argument. The population regression is

$$e(w) = P(\text{entry} \mid w) = 1 - F(0.2 + 0.1w).$$

The assignment law identifies this regression almost everywhere on its support. Absolute continuity of F makes e continuous, so its values at one and four are determined by the population law as well. This is a population identification statement, not exact finite-sample estimation at continuously assigned deployment values.

Under the stable threshold response, with the same V under both deployments, $Y_4 \geq Y_1$ for each district. Hence $\{Y_1 = 1\} \subseteq \{Y_4 = 1\}$ and

$$P(Y_1 = 0 \mid Y_4 = 1) = \frac{P(Y_4 = 1) - P(Y_1 = 1)}{P(Y_4 = 1)} = \frac{e(1) - e(4)}{1 - e(4)}.$$

For the uniform-type illustration, $e(1) = 0.7$ and $e(4) = 0.4$, giving 0.50. This evaluates the same target as PC_E in the original population where deployment four is factual. It does not require the assignment law to place positive probability on receiving exactly four; $e(4)$ is fixed by continuity. Any implementation with the same regime marginals and the stated unit-level monotonicity yields this ratio; random assignment alone does not imply monotonicity.

The replication script retains a 600-district simulation with seed 2026 and a fitted probit as a finite-sample diagnostic. It concerns this uniform-type illustration, not either account in Figure 3.

A.8.5 Continuous-analogue classification

Continuous V generates uncountably many type-indexed loci in the Harsanyi representation. This is therefore an analogue, not a literal application of Definition 4, which requires a declared finite family $\mathcal{H}$.

Under $d : w = w_{\text{small}}$, varying F below the observed cutpoint among E -districts changes arrival mass along the counterfactual threshold margin; the threshold rule and off-margin structure remain fixed. The variation is exposed because

$$P_d(\text{entry}) = 1 - F(0.3),$$

which equals 0.45 for F_1 and 0.95 for F_2 . These entry rates are not the conditional attribution values $1/12$ and $11/12$. The stable-type coupling is the residual implementation: separate-regime marginals do not expose this maintained connection used to evaluate PC_E .